PEARSON EDEXCEL INTERNATIONAL A LEVEL PHYSICS · WPH14

Unit 4: Further Mechanics, Fields and Particles

Revision notes, spec point by spec point (spec points 81–124)

How to use these notes. Every section starts with a spec-point badge and a short summary of what that spec point asks, so you always know which point you are revising. The header at the top of each page names the spec point(s) on that page.

BEYOND THE SPEC, BUT EXAMINED

Content the specification does not spell out, but that past-paper mark schemes have rewarded.

EXAM TIP

How to pick up the marks.

MARK-SCHEME LANGUAGE

Words and phrases mark schemes credit.

COMMON MISTAKE

What examiners keep seeing go wrong.

Contents

Topic 1 – Further Mechanics

81 Impulse and change of momentum

82 Core Practical 9: force and change of momentum

83 Conservation of linear momentum in two dimensions

84 Core Practical 10: analysing collisions with ICT

85 Elastic and inelastic collisions

86 Kinetic energy and momentum: $E_k = p^2/2m$

87 Angular displacement: radians and degrees

88 Angular velocity, speed and period

89 Centripetal acceleration and its derivation

90 A resultant force is needed for circular motion

91 Centripetal force equations

Topic 2 – Electric and Magnetic Fields

92 Electric fields as force fields

93 Electric field strength: $E = F/Q$

94 Coulomb's law: the force between two charges

95 Field strength of a point charge

96 Electric field and electric potential

97 Uniform field between parallel plates: $E = V/d$

98 Potential in a radial field

99 Field lines and equipotentials

100 Capacitance: $C = Q/V$

101 Energy stored by a capacitor

102 Charging and discharging curves; the time constant RC

103 Core Practical 11: capacitor charge and discharge

104 Exponential discharge equations

105 Magnetic flux density, flux and flux linkage

106 Force on a charge moving in a magnetic field

107 Force on a current-carrying conductor

108 E.m.f. from relative motion of a coil and a magnet

109 E.m.f. from a changing current in a linked coil

110 Faraday's and Lenz's laws

Topic 3 – Nuclear and Particle Physics

111 Nucleon number and proton number

112 Alpha scattering and the nuclear model of the atom

113 Thermionic emission and accelerating electrons

114 Particle accelerators and detectors

115 Radius of a charged particle's path: $r = p/BQ$

116 Conservation laws and particle tracks

117 Why high energies are needed to probe nucleons

118 Creation and annihilation of matter and antimatter

119 Energy and mass units: MeV, GeV, MeV/c^2, GeV/c^2

120 Relativistic increase in particle lifetime

121 The standard quark-lepton model

122 Particles and antiparticles

123 Conservation of charge, baryon number and lepton number

124 Writing and interpreting particle equations

Topic 1 – Further Mechanics

SPEC 81

Impulse and change of momentum

WHAT THIS SPEC POINT ASKS

Use impulse = FΔt = Δp, which is Newton's second law written in terms of momentum.

81 Force is the rate of change of momentum

- Momentum p = mv (unit kg m s^{-1} or N s). It is a **vector**, so its direction matters.
- Newton's second law in its general form: the **resultant force** equals the rate of change of momentum.

$$F = \Delta p / \Delta t \quad \text{so} \quad F\Delta t = \Delta p = mv - mu$$

- F = average **resultant** force (N), Δt = time the force acts (s), m = mass (kg), u and v = initial and final velocity (m s^{-1}).
- **Impulse** = FΔt. Its unit is the **N s**, which is the same as kg m s^{-1}. Impulse points the same way as the force.
- If the mass is constant, FΔt = m(v − u) becomes F = ma, the form you met at AS.

81 Force–time graphs

- If the force changes during the impact, the **area under a force–time graph** is the impulse, so it equals the change in momentum.
- For a non-constant force, FΔt uses the **average** force. The peak force is bigger.

81 Why longer contact times mean smaller forces

For a fixed change in momentum, FΔt is fixed, so a **longer** collision time gives a **smaller** average force. Crumple zones, airbags, seat belts, helmets, crash mats and bending your knees on landing all work this way: they **deform**, which **increases the time** of the collision, so for the same Δp the force is **reduced**.

WORKED EXAMPLE: BALL REBOUNDING FROM A WALL

A 0.058 kg tennis ball hits a wall at 25 m s^{-1} and rebounds at 18 m s^{-1}. Contact lasts 4.0 ms. Find the average force on the ball.

Take **away from the wall as positive**: u = −25 m s^{-1}, v = +18 m s^{-1}.

Δp = m(v − u) = 0.058 × (18 − (−25)) = 0.058 × 43 = 2.49 N s

F = Δp ÷ Δt = 2.49 ÷ 4.0 × 10^{-3} = **620 N** (2 s.f.), directed away from the wall.

If you forget the sign change you get 0.058 × 7 = 0.41 N s, which is wrong. A rebound always **adds** the speeds.

MARK-SCHEME LANGUAGE

- "Use of FΔt = Δp" and "use of Δp = mΔv" are separate method marks.
- Cushioning explanations need three linked steps: "material **deforms**" → "**time** of collision **increases**" → "since FΔt = Δp (and Δp is the same), the **force is reduced**". Name the change in momentum.
- Newton's third law pairs: "equal and opposite forces on different objects". For a hovering helicopter, the rotor pushes air down, the air pushes the rotor up; "resultant force is zero, so no acceleration".

Often asked as: force from a momentum change, the area under an F–t graph (MCQ), why a rubber strip or helmet reduces the force, a ball bouncing back (Δp = 2mv for equal speeds), unit of impulse (N s).

COMMON MISTAKE

Saying the force is smaller "because the change in momentum is smaller". Δp is the **same**; it is the **time** that is longer.
Explaining with "energy absorbed" or "friction" but never mentioning time.
Forgetting the sign change on a rebound. Write down your positive direction first.

BEYOND THE SPEC, BUT EXAMINED

- **Use the resultant force.** If the weight also acts during the impulse (a jump, a landing), FΔt = Δp applies to the **resultant** force: e.g. contact force − weight (WPH14 Jun 2022).
- **Fluid streams.** For air or water hitting a surface, mass per second = ρAv. The force needed to stop it is (mass per second) × v. Example: a water jet of area 2.0×10^{-4} m^2 at 15 m s^{-1} ($\rho = 1000$ kg m^{-3}) hits a wall and stops. Mass per second = $1000 \times 2.0 \times 10^{-4} \times 15 = 3.0$ kg s^{-1}, so F = 3.0 × 15 = **45 N** (WPH14 Oct 2024).
- **Compare with a threshold.** Work out an impulse or force and compare it with a stated value, then say whether a claim holds (WPH14 Jan 2026).

SPEC 82

Core Practical 9: force and change of momentum

WHAT THIS SPEC POINT ASKS

Carry out an experiment to investigate how the force on an object is related to its change of momentum.

82 Principle

A constant resultant force F acting for a time Δt should give a change in momentum Δp = FΔt. If you measure Δp for different times (or different forces), a graph of Δp against Δt is a **straight line through the origin with gradient F**.

CORE PRACTICAL 9: METHOD

Apparatus: dynamics trolley with a card of known length on top, runway, two light gates connected to a data logger (or timer), pulley, string, slotted masses on a hanger, electronic balance, metre rule.

1. Measure the mass of the trolley M and the hanging mass m with the balance.
2. **Compensate for friction:** tilt the runway slightly until the trolley, given a small push, rolls at constant speed (equal times at both light gates).
3. Attach the string over the pulley to the hanger. The accelerating force is the weight of the hanger, F = mg.
4. Release the trolley from rest. The light gates record the time for the card to pass each gate (speed = card length ÷ time) and the time Δt between the gates.
5. Repeat for several gate separations (or several hanging masses), three times each, and find means.
6. Calculate $\Delta p = (M + m)(v_2 - v_1)$ for each run and FΔt = mgΔt.

82 Variables

Variable	
Independent	time between the gates Δt (or the accelerating force mg)
Dependent	change in momentum Δp, from the two speeds
Control	total mass (M + m), card length, friction compensation, release point

82 Processing the results

- The force mg accelerates **both** the trolley and the hanging mass, so use the **total** mass:

$$mg\Delta t = (M + m)(v_2 - v_1)$$

- Plot Δp (y-axis) against Δt (x-axis). Gradient = force. Compare it with mg. Or plot Δp against FΔt: gradient should be 1.

WORKED EXAMPLE: CHECKING FΔT = ΔP

Trolley M = 0.800 kg, hanging mass m = 0.050 kg, so total mass 0.850 kg and F = 0.050 × 9.81 = 0.491 N.
Speeds at the gates: v_1 = 0.500 m s^{-1}, v_2 = 1.000 m s^{-1}. Time between gates Δt = 0.87 s.
Δp = 0.850 × (1.000 − 0.500) = **0.43 N s** (2 s.f.)
FΔt = 0.491 × 0.87 = **0.43 N s** (2 s.f.)
The two agree, so the change in momentum equals the impulse.

82 Errors and improvements

- **Friction** not fully compensated: the resultant force is less than mg, so Δp is smaller than FΔt (a systematic error).
- Using only the trolley's mass instead of M + m gives a gradient that is too small.
- Card not perpendicular to the beam, or a card that is too short, gives poor speed readings. Measure the card length with vernier calipers.
- Repeat and average; discard runs where the trolley swerves or touches a gate.
- **Alternative method:** let a trolley hit a **force sensor** on a data logger. The area under the force–time graph (impulse) is compared with m(v − u) from light gates before and after the collision.

EXAM TIP

This practical has not been set as a "describe the method" question so far. It turns up as data: light-gate speeds turned into momentum, a collision time, or an area under an F–t graph. Make sure you can turn a card length and a gate time into a speed, then into momentum.

COMMON MISTAKE

Writing the equation as m t = Mg(v_2 − v_1) (seen in some revision sources). The force is the **hanging weight** mg; the mass being accelerated is (**M + m**). Get the units of each side to match: N s on both.

SPEC 83 Conservation of linear momentum in two dimensions

WHAT THIS SPEC POINT ASKS

Apply conservation of linear momentum to collisions and explosions in two dimensions.

83 The principle

- **Total momentum before = total momentum after**, provided **no resultant external force** acts (a closed system).
- Momentum is a vector, so in 2-D it is conserved **separately in each of two perpendicular directions**.
- Weight and the normal contact force still act on pucks on a table, but they cancel. Friction is the force you have to assume is negligible.

83 Method for a 2-D collision

1. Choose axes: x along the original direction of motion, y perpendicular to it.
2. Resolve every momentum after the collision: p cos θ along x and p sin θ along y, with θ measured from the x-axis. Check which line each angle is measured from.
3. **y-direction:** total y-momentum before is usually **zero**, so the y-components after must cancel. This often gives the first unknown.
4. **x-direction:** total x-momentum before = sum of x-components after.
5. Combine components with **Pythagoras** (magnitude) and **tan θ = p_y / p_x** (direction).

You can also draw a **vector diagram**: momentum vectors drawn to scale, tip to tail. If momentum is conserved, the vectors after add to the vector before, so the diagram **closes into a triangle**.

WORKED EXAMPLE: PUCK COLLISION

Puck A (0.20 kg) moving at 3.0 m s^{-1} hits stationary puck B (0.30 kg). Afterwards A moves at 1.5 m s^{-1} at 50° to its original line. Find B's velocity.
Momentum before: x: 0.20 × 3.0 = 0.60 N s; y: 0.
A after: x: 0.20 × 1.5 cos 50° = 0.193 N s; y: 0.20 × 1.5 sin 50° = 0.230 N s.
x: 0.60 = 0.193 + p_{Bx} → p_{Bx} = 0.407 N s
y: 0 = 0.230 − p_{By} → p_{By} = 0.230 N s (on the other side of the line)
$p_B = \sqrt{(0.407^2 + 0.230^2)}$ = 0.468 N s, so v_B = 0.468 ÷ 0.30 = **1.6 m s^{-1}**
Angle: tan θ = 0.230 ÷ 0.407 → θ = **29°** to the original line, on the opposite side from A.

83 Explosions and recoil

- If everything starts at rest, total momentum is zero, so the fragments' momenta add to zero. Two fragments move off in **exactly opposite** directions with **equal-sized momenta**: $m_1v_1 = m_2v_2$.
- In a radioactive decay with three products (or a recoiling nucleus and two particles), the three momenta form a **closed triangle**.

MARK-SCHEME LANGUAGE

- Condition: "total momentum before = total momentum after, **if no (resultant) external force** acts" (or "in a closed system").
- For pucks on ice or air tables: "no / negligible friction", so "**no resultant external force**". "Reduces friction" alone is not enough.
- Marks: "use of p = mv", "use of trigonometry to find a component", "conservation in one direction", then "in both directions". A final comparison mark wants "momentum before = momentum after (in both directions)".
- Vector diagram: "vectors drawn to scale", "end to end", "with arrows", "a **closed triangle**, so momentum is conserved".

Often asked as: pucks, curling stones, cars at a junction, ships, alpha particles scattering off nuclei, decay with a recoiling nucleus.

COMMON MISTAKE

Conserving **velocity** instead of momentum, or calculating kinetic energy where momentum was wanted (almost a fifth of one cohort did this).
Swapping sin and cos. If θ is measured from the x-axis, the x-component uses cos θ.
Stopping after the x-equation. A 2-D question needs both directions to finish.
Cancelling the masses without showing it.

BEYOND THE SPEC, BUT EXAMINED

- **Equal masses, elastic collision, one at rest:** the two bodies move off **at 90° to each other**. Questions ask you to test this from measured angles (WPH14 Jun 2024).
- **Alpha tracks with a recoil:** a short gap at the start of a track shows the nucleus recoiled between two decays (WPH14 Oct 2020).

SPEC 84

Core Practical 10: analysing collisions with ICT

WHAT THIS SPEC POINT ASKS

Use ICT (video and analysis software) to analyse collisions between small spheres, such as ball bearings on a table top.

84 The idea

Film a collision between two spheres from above, then use video-analysis software (e.g. Tracker) to find each sphere's velocity before and after, frame by frame. Then test conservation of momentum in two directions, and test whether kinetic energy is conserved.

CORE PRACTICAL 10: METHOD

1. Measure the mass of each sphere on an electronic balance.
2. Place a metre rule (or a scale of known length) flat in the field of view, at the same level as the spheres, for calibration.
3. Mount the camera (or phone) **directly above** the table, looking straight down, so the view is perpendicular to the surface (no parallax).
4. Roll one sphere (e.g. down a short ramp for a repeatable speed) into a stationary sphere so that they hit off-centre and move off at angles.
5. Load the video into the software. Set the scale from the rule and the time between frames from the frame rate. Mark each sphere's position frame by frame.
6. Read off the velocity components (or speed and angle) before and after. Repeat for different impact angles.

Low-tech alternative: put carbon paper under paper on the table so the spheres leave tracks; measure angles with a protractor. Speeds can be found from distance travelled between known times.

84 Analysis

- Calculate p = mv for each sphere and resolve into components along and perpendicular to the original direction (as in 83).
- Check: x-momentum before ≈ x-momentum after; y-components after ≈ cancel.
- Check kinetic energy before and after to decide elastic or inelastic (see 85).

- Compare each measured value with the calculated one and say if they agree **within experimental uncertainty**. A difference of about 1° in an angle is "consistent".

84 Errors

Source	Type	Fix
Camera not directly above, or scale not at the spheres' height	systematic (parallax / scale)	mount camera vertically; scale in the same plane
Frame rate too low; blurred images	random	higher frame rate, faster shutter, good lighting
Friction and rolling (some energy goes into rotation)	systematic	smooth level surface; hard spheres
Spheres not hitting cleanly	random	guide ramp; repeat runs

MARK-SCHEME LANGUAGE

- Measuring angles: "film from directly **above**", "measure angles on playback with a **protractor**" (or motion-tracking software), or "carbon paper to record tracks, then protractor".
- Comparison: "compare the student's measured value with the calculated value", then "**conclusion**: consistent / not consistent".

Often asked as: how to measure the angles; use given angles and speeds to test conservation of momentum; is the collision elastic?

EXAM TIP

State the conclusion explicitly and give both numbers with units. "41.5° compared with a measured 43°, which agrees within the uncertainty of a protractor reading, so momentum is conserved."

SPEC 85 Elastic and inelastic collisions

WHAT THIS SPEC POINT ASKS

Decide whether a collision is elastic or inelastic.

85 Momentum always, kinetic energy sometimes

- In **every** collision (with no external force), **momentum is conserved** and **total energy is conserved**.
- **Elastic collision:** total **kinetic energy** is also conserved (Ek before = Ek after).
- **Inelastic collision:** some kinetic energy is transferred to other stores (internal energy/heating, sound, deforming the objects), so Ek after < Ek before.
- When objects **stick together** the collision is inelastic, and the most Ek is lost that momentum conservation allows.
- In an **explosion**, Ek increases (energy comes from a chemical or nuclear store), but momentum is still conserved.

85 How to test

1. Calculate Ek = ½mv^2 (or p^2/2m) for **every** body before the collision and add them.
2. Do the same after.
3. **Compare** the two totals, with units, and state the conclusion.

WORKED EXAMPLE: WAS THE PUCK COLLISION ELASTIC?

Using the puck collision in 83 (A: 0.20 kg, 3.0 m s^{-1} → 1.5 m s^{-1}; B: 0.30 kg, 0 → 1.56 m s^{-1}).
Ek before = ½ × 0.20 × 3.0^2 = 0.90 J
Ek after = ½ × 0.20 × 1.5^2 + ½ × 0.30 × 1.56^2 = 0.225 + 0.364 = 0.59 J
0.59 J < 0.90 J, so kinetic energy is **not** conserved: the collision is **inelastic**. About 0.31 J went into heating and sound.

MARK-SCHEME LANGUAGE

- Marks: "correct Ek for one object", "all Ek values correct", "**conclusion consistent with the calculated values**".
- "Elastic because total Ek before = total Ek after"; "not elastic because Ek before ≠ Ek after" (or "Ek before greater than Ek after").
- Near-equal answers are accepted as elastic: 9.62 × 10^{-3} J and 9.75 × 10^{-3} J "may be considered approximately equal".

Often asked as: the last part of nearly every 2-D collision question; MCQ "momentum conserved, kinetic energy not conserved".

COMMON MISTAKE

Leaving out the comparison sentence: it is a mark on its own.
Forgetting a body that was already moving before the collision, or adding only one of the energies after.
No unit (J) anywhere in the working.
Saying "energy is lost" in an inelastic collision. Total energy is conserved; **kinetic** energy is not.

BEYOND THE SPEC, BUT EXAMINED

- Using the elastic condition (Ek before = Ek after) as an extra equation to find an unknown speed (WPH14 Jan 2026).
- Applying the test to particles, e.g. an alpha particle colliding with a proton, with energies around 10^{-12} J (WPH14 Jun 2023).

SPEC 86

Kinetic energy and momentum: Ek = $p^2/2m$

WHAT THIS SPEC POINT ASKS

Derive and use Ek = $p^2/2m$ for a particle moving much slower than light.

86 Derivation

Start from Ek = ½mv^2 and p = mv, so v = p/m:

$$E_k = \tfrac{1}{2}\, m\, (p/m)^2 = p^2 / 2m \quad \text{and so} \quad p = \sqrt{(2mE_k)}$$

- Ek in J, p in kg m s^{-1} (N s), m in kg.
- **Non-relativistic only.** If ½mv^2 gives a speed near or above c (3.00 × 10^8 m s^{-1}), the formula is not valid.
- Use it whenever you know the momentum (from conservation, from r = p/BQ, or from λ = h/p) and want the energy, or the other way round.

86 Quick ratio results (for MCQs)

- Same momentum: Ek ∝ 1/m, so the **lighter** body has more kinetic energy (a recoiling nucleus has less Ek than the alpha particle).
- Same mass: Ek ∝ p^2. Halve the momentum and Ek falls to a **quarter**.

WORKED EXAMPLE: ELECTRONS WITH A CHOSEN DE BROGLIE WAVELENGTH

Electrons for a diffraction experiment need $\lambda = 1.0 \times 10^{-10}$ m. What accelerating p.d. is needed?
$p = h/\lambda = 6.63 \times 10^{-34} \div 1.0 \times 10^{-10} = 6.63 \times 10^{-24}$ kg m s^{-1}
$Ek = p^2/2m = (6.63 \times 10^{-24})^2 \div (2 \times 9.11 \times 10^{-31}) = 2.41 \times 10^{-17}$ J
$eV = Ek \rightarrow V = 2.41 \times 10^{-17} \div 1.60 \times 10^{-19} =$ **150 V** (2 s.f.)

WORKED EXAMPLE: MOMENTUM OF A PROTON

A proton has Ek = 5.0 MeV. $Ek = 5.0 \times 10^6 \times 1.60 \times 10^{-19} = 8.0 \times 10^{-13}$ J.
$p = \sqrt{(2 \times 1.67 \times 10^{-27} \times 8.0 \times 10^{-13})} =$ **5.2×10^{-20} kg m s^{-1}**

MARK-SCHEME LANGUAGE

- "Use of $Ek = p^2/2m$ **Or** use of $Ek = \frac{1}{2}mv^2$ and $p = mv$": both routes score.
- A separate mark for converting eV (or MeV) to J.
- "Show each step and say which equation" in a de Broglie show-that.

Often asked as: combined with r = p/BQ to find B, combined with λ = h/p to find V, mass of an ion from p and Ek, Ek after a collision from momenta, MCQ on how Ek changes when p changes.

COMMON MISTAKE

Using the electron mass for a proton (or the other way round).
Dropping the "M" in MeV, or dividing by 1.6×10^{-19} when you should multiply.
The formula $p = \sqrt{(2mEk)}$ is not in every formula list, so learn the derivation.

BEYOND THE SPEC, BUT EXAMINED

- **de Broglie wavelength** λ = h/p (from Unit 2) is combined with $Ek = p^2/2m$ and Ek = eV almost every year (WPH14 Jan 2024, Jan 2025).
- Showing that $\frac{1}{2}mv^2$ gives v > c for a 23 MeV electron, so a relativistic treatment is needed (WPH14 Jun 2021).

SPEC 87 Angular displacement: radians and degrees

WHAT THIS SPEC POINT ASKS

Express angles in radians and degrees, and convert between them.

87 The radian

- **Angular displacement** θ is the angle turned through about the centre.
- One **radian** is the angle at the centre of a circle when the arc length equals the radius. In general:

$$\theta \text{ (rad)} = \text{arc length } s \,/\, \text{radius } r$$

- A full circle: arc = 2πr, so **360° = 2π rad**. Half a turn = π rad; a quarter turn = π/2 rad; 1 rad ≈ 57.3°.

Degrees	30°	45°	60°	90°	180°	360°
Radians	π/6	π/4	π/3	π/2	π	2π

87 Converting

$$\text{angle in rad} = \text{angle in degrees} \times 2\pi / 360 \qquad \text{angle in degrees} = \text{angle in rad} \times 360 / 2\pi$$

WORKED EXAMPLE: FROM DEGREES TO ANGULAR SPEED

A ride turns through 160° in 2.2 s. Find its angular speed.
θ = 160 × 2π/360 = 2.79 rad
ω = θ/t = 2.79 ÷ 2.2 = **1.3 rad s^{-1}**
Other route: one full turn takes 360/160 × 2.2 = 4.95 s = T, so ω = 2π/T = 1.3 rad s^{-1}.
Going back the other way: 1.3 rad = 1.3 × 360/2π = 74°.

MARK-SCHEME LANGUAGE

- Both routes score: convert to radians then ω = Δθ/Δt, or find the period by proportion then ω = 2π/T.
- Unit "rad s^{-1}" (or "radian s^{-1}").

Often asked as: a fraction of a turn (quarter circle = 2πr/4 of arc), time to turn through a given angle, MCQ conversions (40° = 40 × 2π/360 rad).

COMMON MISTAKE

Not turning 90° into a quarter of a circle, or applying an extra factor of 4.
Multiplying a value that is already in radians by π again.
Leaving the calculator in degree mode when taking sin or cos of an angle in radians.
Units of ω as m s^{-1} or "degrees per second".

SPEC 88 Angular velocity, speed and period

WHAT THIS SPEC POINT ASKS

Understand angular velocity and use v = ωr and T = 2π/ω.

88 Definitions

- **Angular velocity** ω is the rate of change of angular displacement: ω = Δθ/Δt, unit **rad s^{-1}**.
- For one full turn, Δθ = 2π in one **period** T, so:

$$\omega = 2\pi / T = 2\pi f \qquad v = \omega r$$

- v = linear (tangential) speed (m s^{-1}), r = radius (m), T = period (s), f = frequency (Hz, revolutions per second).
- v = ωr comes from s = rθ: dividing by t gives v = rω.
- All points on a rigid rotating object have the **same ω**, but points further out have a **larger v**.

WORKED EXAMPLE: A WASHING-MACHINE DRUM

The drum spins at 1400 revolutions per minute and has a diameter of 0.50 m. Find ω and the speed of the drum wall.

f = 1400 ÷ 60 = 23.3 Hz (revolutions per **second**)

ω = 2πf = 2π × 23.3 = **147 rad s^{-1}**

r = 0.50 ÷ 2 = 0.25 m, so v = ωr = 147 × 0.25 = **37 m s^{-1}**

MARK-SCHEME LANGUAGE

- "Use of ω = 2π/T" and "use of v = ωr" are separate method marks.
- ω needs the unit rad s^{-1} (s^{-1} is accepted in a derivation).

Often asked as: ω from a period or rpm, v from ω, time for part of a turn, the first step of a centripetal force calculation.

COMMON MISTAKE

Using the **diameter** as the radius: the most-quoted error in the examiner reports.

Writing ω = vr instead of v = ωr.

Forgetting to change revolutions per **minute** to per second.

Leaving the answer in terms of π.

BEYOND THE SPEC, BUT EXAMINED

A coil rotating at ω in a magnetic field gives a peak e.m.f. of BANω. Halving ω doubles the period and halves the peak e.m.f. (WPH14 Jan 2020; see 110).

SPEC 89

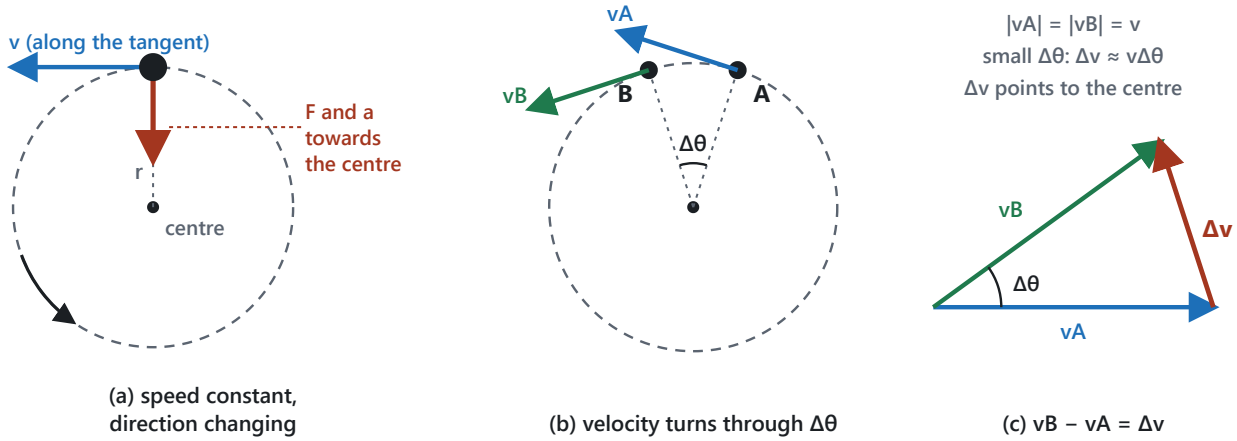
Centripetal acceleration and its derivation

WHAT THIS SPEC POINT ASKS

Use vector diagrams to derive $a = v^2/r = r\omega^2$, and use these equations.

89 Why there is an acceleration

An object moving in a circle at **constant speed** is still **accelerating**, because its **velocity** (a vector) keeps changing **direction**. The acceleration points **towards the centre**: it is called **centripetal** acceleration.



(a) speed constant, direction changing

(b) velocity turns through Δθ

(c) vB − vA = Δv

(a) The velocity is along the tangent; the resultant force and the acceleration point to the centre. (b) In time Δt the velocity turns through Δθ. (c) The change in velocity Δv closes the triangle and points towards the centre.

89 The derivation (4 marks, often asked)

1. The object moves from A to B in a short time Δt. The radius turns through angle Δθ, and so does the velocity: v_A and v_B have the same size v but differ in direction by Δθ.
2. Draw the vector triangle: $v_B - v_A = \Delta v$ (v_A and v_B from a common point; Δv joins the tip of v_A to the tip of v_B).
3. For a **small** angle, the triangle's short side is almost an arc of radius v, so **Δv ≈ vΔθ** (Δθ in radians).
4. Acceleration $a = \Delta v/\Delta t = v(\Delta\theta/\Delta t) =$ **vω**.
5. Substitute $\omega = v/r$ or $v = \omega r$:

$$a = v^2 / r = r\omega^2$$

- a = centripetal acceleration (m s^{-2}), v = speed (m s^{-1}), r = radius (m), ω = angular velocity (rad s^{-1}).
- As Δθ → 0, Δv becomes perpendicular to v, so a points along the radius **towards the centre**.

WORKED EXAMPLE: A CAR ON A ROUNDABOUT

A car goes round a curve of radius 40 m at 15 m s^{-1}.
$a = v^2/r = 15^2 \div 40 =$ **5.6 m s^{-2}** towards the centre of the curve.
$\omega = v/r = 15 \div 40 = 0.375$ rad s^{-1}; check: $r\omega^2 = 40 \times 0.375^2 = 5.6$ m s^{-2} ✓

MARK-SCHEME LANGUAGE

The four marks:

- "Correct **vector diagram** showing the velocity change" (v_A, v_B, Δv and the angle Δθ, all with arrows).
- "**Small angle**, so Δθ = Δv/v" (or "sin θ ≈ θ", or "arc ≈ chord").
- "Use of Δθ/Δt = ω and v = rω".
- "Algebra to show $a = v^2/r$" (with $a = \Delta v/\Delta t$).

A similar-triangles route also scores: Δv/v = Δs/r with Δs/Δt = v.

Often asked as: the derivation on its own; MCQ on which arrow is Δv; using $a = v^2/r$ inside a force question.

COMMON MISTAKE

Diagram with no arrows, or with arrows that don't fit $v_B - v_A = \Delta v$.
Not saying **why** the angle must be small (a 90° angle is not "small").
Missing the step $\Delta\theta = \omega\Delta t$.

SPEC 90

A resultant force is needed for circular motion

WHAT THIS SPEC POINT ASKS

A resultant (centripetal) force is needed to produce and keep up circular motion.

90 Newton's laws applied to circles

- **First law:** with no resultant force, an object moves in a **straight line** at constant speed. To make it follow a curve, something must push or pull it sideways.
- **Second law:** the object accelerates towards the centre, so the **resultant force** must also point **towards the centre**. This resultant is called the **centripetal force**.
- The force is **perpendicular to the velocity**, so it does **no work**: the speed stays constant, only the direction changes.
- If the force stops, the object carries on **along the tangent**, not outwards.

90 "Centripetal force" is not an extra force

It is the **name for the resultant** of the real forces. Always say what provides it:

Situation	What provides the centripetal force
Ball whirled on a string, hammer throw	tension in the string/wire
Car on a flat bend	friction between tyres and road
Car on a banked track, aircraft banking	horizontal component of the normal reaction / lift
Conical pendulum	horizontal component of the tension
Satellite, Moon	gravitational pull of the Earth
Electron orbiting a nucleus	electrostatic (Coulomb) attraction
Charged particle in a magnetic field	magnetic force BQv
Passenger in a turning car or ride	push from the seat, door or wall

90 Vertical circles

- At the **top**, weight and the push (or tension) both point to the centre: $\mathbf{mg + R = mv^2/r}$.
- At the **bottom**, the push (or tension) points up and the weight down: $\mathbf{R - mg = mv^2/r}$, so R is largest at the bottom.
- The minimum speed at the top is when R (or T) just reaches **zero**: $mg = mv^2/r$, so $\mathbf{v_{min} = \sqrt{(gr)}}$.

MARK-SCHEME LANGUAGE

Passenger who "feels pushed outwards" (three marks):

- "By Newton's first law the passenger would continue in a **straight line**."
- "A (centripetal) force acts **towards the centre** of the circle."
- "This force is exerted **by the car / seat on the passenger**; there is **no outward force**."

Free-body diagrams: label weight "W, mg or gravitational force", tension "T", normal reaction "R" or "N". Do not add a "centripetal force" arrow.
Thrown from a circle: the object leaves "**at a tangent**" ("shoots away" scores nothing).

Often asked as: free-body diagrams for banked tracks, conical pendulums, vertical circles; why the reaction force changes round a loop; why water stays in a bucket at the top.

COMMON MISTAKE

Drawing "centripetal force" as an extra arrow on a free-body diagram.
Inventing a "centrifugal force" as the Newton's third law pair of the centripetal force. The third-law partner of the seat pushing you is **you pushing the seat**.
Saying only "at right angles to the motion" without "towards the centre".
Labelling forces with vague letters like F or G.

BEYOND THE SPEC, BUT EXAMINED

- Variation of the normal reaction around a vertical loop, read from an accelerometer trace on a roller coaster (WPH14 Jun 2023, 6-mark).
- Objects on a turntable: friction ∝ mass and centripetal force ∝ mass, so the radius at which things slide does **not** depend on mass (WPH14 Jun 2025).

SPEC 91

Centripetal force equations

WHAT THIS SPEC POINT ASKS

Use $F = ma = mv^2/r = mr\omega^2$ for the centripetal force.

$$F = ma = mv^2 / r = mr\omega^2$$

- F = resultant (centripetal) force (N), m = mass (kg), v = speed (m s^{-1}), r = radius (m), ω = angular velocity (rad s^{-1}).
- Combine with $\omega = 2\pi/T$ when you are given a time for one turn.

91 Method for angled forces

1. Draw a free-body diagram with the real forces only.
2. **Resolve vertically:** no vertical acceleration, so the vertical components balance the weight.
3. **Resolve horizontally:** the horizontal component is the centripetal force mv^2/r.
4. Divide one equation by the other to cancel the unknown force. For a banked track or a banking aircraft this gives **$\tan\theta = v^2/rg$**.
5. For a pendulum on a string of length l at angle θ to the vertical, the radius is **$r = l \sin\theta$**.

WORKED EXAMPLE: CONICAL PENDULUM

A 0.20 kg bob on a 0.80 m string moves in a horizontal circle with the string at 30° to the vertical. Find the tension and ω.

Vertically: $T \cos 30° = mg \rightarrow T = 0.20 \times 9.81 \div \cos 30° =$ **2.3 N**

Radius: $r = l \sin \theta = 0.80 \times \sin 30° = 0.40$ m

Horizontally: $T \sin 30° = mr\omega^2 \rightarrow \omega^2 = (2.27 \times 0.5) \div (0.20 \times 0.40) = 14.2 \rightarrow \omega =$ **3.8 rad s^{-1}**

WORKED EXAMPLE: BANKED TRACK

A track of radius 120 m is designed so cars at 25 m s^{-1} need no friction. Find the banking angle.

$R \cos \theta = mg$ and $R \sin \theta = mv^2/r$. Divide: $\tan \theta = v^2/rg = 25^2 \div (120 \times 9.81) = 0.531$

$\theta =$ **28°**

WORKED EXAMPLE: VERTICAL CIRCLE

A 0.50 kg ball on a string of radius 0.90 m moves at 4.0 m s^{-1} at the bottom of a vertical circle. Find the tension there.

$T - mg = mv^2/r \rightarrow T = 0.50 \times 9.81 + 0.50 \times 4.0^2 \div 0.90 = 4.91 + 8.89 =$ **14 N**

Minimum speed at the top ($T = 0$): $v = \sqrt{(gr)} = \sqrt{(9.81 \times 0.90)} =$ **3.0 m s^{-1}**

MARK-SCHEME LANGUAGE

- Separate marks for "**resolve vertically**" and "**resolve horizontally**", then "use of $F = mv^2/r$", then "use radius = $l \sin \theta$".
- Steeper bank than the design angle: "vertical component of lift is now greater than the weight, so the plane accelerates upwards".
- Comparison with a limit: "T = 31 N, less than the 35 N breaking tension, so the wire does not break".

Often asked as: banked tracks and aircraft, conical pendulums, wire tension in a hammer throw (Pythagoras of mv^2/r and mg), minimum speed at the top of a vertical circle, reading a maximum force off a graph to find r.

COMMON MISTAKE

Taking the string length as the radius, or using tan where sin is needed for r.

Multiplying by cos θ when you should divide ($T = mg/\cos \theta$, not $mg \cos \theta$).

Jumping straight to $\tan \theta = v^2/rg$ without showing the two components: derivations need both equations.

Mixing up which component is vertical and which horizontal.

EXAM TIP

In a "show that" for circular motion, give your answer to one more significant figure than the value printed in the question. A typical chain: $\omega = 2\pi/T \rightarrow v = \omega r \rightarrow F = mv^2/r$.

Topic 2 – Electric and Magnetic Fields

SPEC 92 Electric fields as force fields

WHAT THIS SPEC POINT ASKS

An electric field is a region where a charged particle experiences a force.

- Every charge sets up an electric field around itself. Any **other** charge placed in that field feels a force.
- **Like** charges repel; **unlike** charges attract.
- The **direction of an electric field** at a point is the direction of the force on a small **positive** test charge placed there. A negative charge feels a force the opposite way.
- So field lines point **away from positive** charges and **towards negative** charges.
- Electric forces act on **charges whether they move or not**. Compare: a gravitational field acts on **mass**, and a magnetic field acts on **moving** charges (and currents).

Field	Acts on	Defined as	Unit
Gravitational	mass	force per unit mass, g = F/m	N kg^{-1}
Electric	charge	force per unit positive charge, E = F/Q	N C^{-1} (= V m^{-1})
Magnetic	moving charge or current	B from F = BIl	T

MARK-SCHEME LANGUAGE

- Definition: "region / area where a **charged particle** experiences a **force**". "Region where a charge feels a force" is enough; "region of force" alone is not.
- Sign of a charge from a diagram: "**negative**, because the field direction is the direction of the force on a **positive** charge" (e.g. Earth's surface, where the fair-weather field points downwards).

Often asked as: one-mark definition; MCQ reading the sign of charges X and Y from arrowed field lines.

COMMON MISTAKE

Reading an arrow as the direction an **electron** (or any negative charge) moves. Field lines show the force on a **positive** charge.

SPEC 93 Electric field strength: E = F/Q

WHAT THIS SPEC POINT ASKS

Electric field strength is defined as E = F/Q; use this equation.

$$E = F / Q$$

- E = electric field strength (N C^{-1}), F = force on the charge (N), Q = charge (C).
- In words: the **force per unit positive charge** at that point.
- E is a **vector**. Its direction is the direction of the force on a positive charge.
- Rearranged, **F = EQ** gives the force on any charge in a known field. For a negative charge, F points opposite to E.

- Once you have F, use the rest of mechanics: a = F/m, work done = F × distance, or combine with the weight.

WORKED EXAMPLE: FORCE AND ACCELERATION OF A PROTON

A proton sits in a uniform field of 5.0×10^3 N C^{-1}.
F = EQ = $5.0 \times 10^3 \times 1.60 \times 10^{-19}$ = **8.0×10^{-16} N**, in the direction of the field.
a = F/m = $8.0 \times 10^{-16} \div 1.67 \times 10^{-27}$ = **4.8×10^{11} m s^{-2}**
An electron in the same field feels the same size of force, but in the **opposite** direction, and its acceleration is about 1800 times larger.

MARK-SCHEME LANGUAGE

- "Use of F = EQ", then "F − W to find the resultant force" if the particle also has weight, then "use of F = ma".
- Field strength units: N C^{-1} **or** V m^{-1} are both accepted.

Often asked as: force on a charged drop or ion, then its acceleration; force between plates, then the work done over a distance.

COMMON MISTAKE

Using V/d as if it were the force. V/d is the **field**; multiply by Q to get the force.
Using the total of two charges for Q when only one charge is in the field.

SPEC 94

Coulomb's law: the force between two charges

WHAT THIS SPEC POINT ASKS

Use $F = Q_1Q_2/4\pi\varepsilon_0 r^2$ for the force between two point charges.

$$F = Q_1Q_2 / 4\pi\varepsilon_0 r^2$$

- F = force between the charges (N), Q_1 and Q_2 = the charges (C), r = separation of their **centres** (m).
- ε_0 = permittivity of free space = 8.85×10^{-12} F m^{-1} (data booklet). The constant $1/4\pi\varepsilon_0 = 8.99 \times 10^9$ N m^2 C^{-2} is accepted too: $F = kQ_1Q_2/r^2$.
- It is an **inverse-square law**: double r and F falls to **a quarter**; triple r and F falls to a ninth.
- Charged **spheres** behave as point charges at their centres.
- Sign: a positive answer means repulsion (like charges); a negative one means attraction. Most questions only want the size.
- The same-sized force acts on **each** charge, in opposite directions (Newton's third law).

WORKED EXAMPLE: TWO SMALL CHARGED SPHERES

Charges of +3.0 nC and +5.0 nC have centres 2.5 cm apart.
F = $(8.99 \times 10^9 \times 3.0 \times 10^{-9} \times 5.0 \times 10^{-9}) \div (0.025)^2$ = $1.35 \times 10^{-7} \div 6.25 \times 10^{-4}$ = **2.2×10^{-4} N**, repulsive.
Convert first: nC → × 10^{-9} C; cm → m.

WORKED EXAMPLE: THE HYDROGEN ATOM

An electron and a proton are 5.3×10^{-11} m apart.
$F = 8.99 \times 10^{9} \times (1.60 \times 10^{-19})^2 \div (5.3 \times 10^{-11})^2 =$ **8.2×10^{-8} N** (attractive)
If this provides the centripetal force: $mv^2/r = F \rightarrow v = \sqrt{(Fr/m)} = \sqrt{(8.2 \times 10^{-8} \times 5.3 \times 10^{-11} \div 9.11 \times 10^{-31})}$
$=$ **2.2×10^{6} m s^{-1}**

MARK-SCHEME LANGUAGE

- "Use of $F = Q_1Q_2/4\pi\varepsilon_0r^2$ (accept $F = kQ_1Q_2/r^2$)". The sign of F is optional.
- For nuclei, a separate mark for the charge: alpha = **2e**, gold nucleus = **79e**, each e = 1.60×10^{-19} C.

Often asked as: force between spheres or particles; separation for a given force; MCQ on F when r doubles (F/4); force on an electron in an atom set equal to mv^2/r.

COMMON MISTAKE

Not squaring r, or squaring the charges (the classic MCQ distractors).
Using the diameter for r, or measuring from the surfaces instead of the centres.
Forgetting the unit on the final answer.

BEYOND THE SPEC, BUT EXAMINED

- **Charged spheres on threads:** the thread makes angle θ with the vertical; resolving gives **tan θ = F/W**, so F = mg tan θ, and then Q from Coulomb's law (WPH14 Jan 2022, Jun 2023).
- **Coulomb force then motion:** F = ma, then a time or distance with suvat, compared with a claim (WPH14 Jun 2024).
- **Atom as circular motion:** Coulomb force = mv^2/r for the electron (WPH14 Jan 2026).

SPEC 95

Field strength of a point charge

WHAT THIS SPEC POINT ASKS

Use $E = Q/4\pi\varepsilon_0r^2$ for the electric field of a point charge.

$$E = Q / 4\pi\varepsilon_0 r^2$$

- E = field strength (N C^{-1}), Q = the charge **making** the field (C), r = distance from its centre (m).
- This comes from E = F/Q with Coulomb's law: put a test charge q at r; $F = Qq/4\pi\varepsilon_0r^2$, so $F/q = Q/4\pi\varepsilon_0r^2$.
- **Radial** field: lines spread out from the charge, so $E \propto 1/r^2$. Double the distance, quarter the field.
- Direction: **away** from a positive charge, **towards** a negative charge.
- A charged conducting **sphere** gives the same field outside it as a point charge at its centre.

95 More than one charge

Fields are **vectors**. Work out each field separately (size from the formula, direction from the sign of the charge), then add them as vectors. On a straight line: same direction → **add**; opposite directions → **subtract**.

WORKED EXAMPLE: FIELD BETWEEN TWO CHARGES

+6.0 nC and −4.0 nC are 0.20 m apart. Find E at the midpoint (0.10 m from each).
From +6.0 nC: $E_1 = 8.99 \times 10^9 \times 6.0 \times 10^{-9} \div 0.10^2 = 5390$ N C^{-1}, pointing **away** from it (towards the −4.0 nC).
From −4.0 nC: $E_2 = 8.99 \times 10^9 \times 4.0 \times 10^{-9} \div 0.10^2 = 3600$ N C^{-1}, pointing **towards** it (also towards the −4.0 nC).
Same direction, so add: E = 5390 + 3600 = **9.0×10^3 N C^{-1}** towards the negative charge.
With two **like** charges the fields at the midpoint point opposite ways and you subtract. Equal like charges give **zero** field at the midpoint: a **neutral point**.

MARK-SCHEME LANGUAGE

- "Adds E due to X to E due to Y" **with the directions considered**.
- Units N C^{-1} or V m^{-1}.

Often asked as: field at a point from one or two charges; the charge on the Earth from the field at its surface; field in a molecule.

COMMON MISTAKE

Subtracting when both fields point the same way (nearly half of one cohort did this).
Using Q_1Q_2 in a field question. A field needs only the **source** charge.
Giving field strength in V instead of V m^{-1}.

BEYOND THE SPEC, BUT EXAMINED

- **Surface charge density** = Q/A with A = $4\pi r^2$ for a sphere (WPH14 Jan 2020).
- **Capacitance of an isolated sphere**, C = $4\pi\varepsilon_0 r$, used with Q = CV to find the charge on a sphere (WPH14 Oct 2023).

SPEC 96

Electric field and electric potential

WHAT THIS SPEC POINT ASKS

Know and understand how electric field strength is related to electric potential.

96 Electric potential

- The **electric potential** V at a point is the **work done per unit positive charge** in bringing a small positive charge from **infinity** to that point. Unit: volt, V = J C^{-1}.
- Potential is a **scalar**: it has no direction, only a sign.
- Potential at infinity is taken as **zero**. Near a positive charge V is positive; near a negative charge V is negative.
- **Potential difference** between two points is the work done per unit charge moving between them. Moving a charge Q through a p.d. ΔV transfers energy:

$$\Delta W = Q\Delta V$$

96 Field = potential gradient

- The field strength is the **rate of change of potential with distance**, with a minus sign:

$$E = -dV/dr \quad \text{(in a uniform field, } E = -\Delta V/\Delta x\text{)}$$

- The minus sign means the field points from **high potential to low potential** (downhill).
- On a graph of V against r: field strength = **minus the gradient**. A steeper graph means a stronger field.
- On a graph of E against r: the **area** under the graph between two distances = the potential difference between them.
- Units check: V m^{-1} = (J C^{-1}) m^{-1} = (N m C^{-1}) m^{-1} = **N C^{-1}**. The two units of E are the same thing.

WORKED EXAMPLE: ENERGY AND FIELD FROM POTENTIALS

An alpha particle (charge 2e) moves from the 40 V equipotential to the 10 V equipotential.
$\Delta W = Q\Delta V = 2 \times 1.60 \times 10^{-19} \times 30 =$ **9.6×10^{-18} J** (= 60 eV) of kinetic energy gained.
If those equipotentials were 3.0 mm apart in a uniform field: $E = \Delta V/\Delta x = 30 \div 3.0 \times 10^{-3} =$ **1.0×10^{4} V m^{-1}**.

MARK-SCHEME LANGUAGE

- Field strength is the "potential gradient"; "potential is the area under the E–r graph" (MCQ explanations).
- Electric potential is a **scalar** (MCQs listing scalars and vectors).
- Showing $V \propto 1/r$ from a diagram: measure r to two equipotentials and show "V × r is constant".

Often asked as: MCQ on area under E–r or gradient of V–r; energy in eV gained between equipotentials.

EXAM TIP

In eV, the energy gained is just (charge in units of e) × (p.d. in volts). An alpha crossing 30 V gains 2 × 30 = 60 eV. Convert to J only at the end.

SPEC 97

Uniform field between parallel plates: E = V/d

WHAT THIS SPEC POINT ASKS

Use E = V/d for the uniform field between parallel plates.

$$E = V / d$$

- E = field strength (V m^{-1}), V = p.d. between the plates (V), d = plate separation (m).
- Why: moving a charge Q from one plate to the other takes work QV = Fd, so F/Q = V/d.
- Between the plates the field is **uniform**: the same size and direction everywhere (except near the edges). The force on a charge is the same wherever it is between the plates.
- The field points from the **positive** plate to the **negative** (or earthed, 0 V) plate.

97 A charge entering the field sideways

An electron entering between horizontal plates at speed u along the plates behaves like a **projectile**:

- **Along** the plates: no force, so **constant speed**: t = length ÷ u.
- **Across** the plates: constant force F = eE, so constant acceleration a = eE/m. Use $s = \frac{1}{2}at^2$ for the sideways deflection.
- The path between the plates is a **parabola**; after leaving the plates it is a straight line.

WORKED EXAMPLE: DEFLECTION OF AN ELECTRON BEAM

Plates 2.5 cm apart have a p.d. of 500 V and are 5.0 cm long. An electron enters midway between them at 2.0×10^7 m s^{-1}, parallel to the plates.
$E = V/d = 500 \div 0.025 = 2.0 \times 10^4$ V m^{-1}
$a = eE/m = (1.60 \times 10^{-19} \times 2.0 \times 10^4) \div 9.11 \times 10^{-31} = 3.51 \times 10^{15}$ m s^{-2}
$t = 0.050 \div 2.0 \times 10^7 = 2.5 \times 10^{-9}$ s
$s = \frac{1}{2}at^2 = 0.5 \times 3.51 \times 10^{15} \times (2.5 \times 10^{-9})^2 = 0.011$ m = **1.1 cm** towards the positive plate.
That is less than the 1.25 cm to the plate, so the electron gets through.

MARK-SCHEME LANGUAGE

- Projectile wording: "resultant force on the electrons in the **vertical** direction, so they **accelerate vertically**; in the **horizontal** direction they have **constant speed**".
- Uniform-field diagrams: "at least three **equally spaced parallel** lines", "touching both plates", "arrow from + to −".
- Charge as a multiple of e: "Q = 5e" or "integer multiple of 1.6×10^{-19} C". A value like 4.97e should be recognised as **5e**.

Often asked as: force on a particle between plates; speed gained crossing the gap (Ek = QV); deflection of an electron beam; Millikan-type drops; charged spheres shuttling between plates.

COMMON MISTAKE

Leaving d in cm or mm.
Using V/d as the force instead of the field.
Applying suvat along the plates as if there were an acceleration that way.

BEYOND THE SPEC, BUT EXAMINED

- **Millikan's oil-drop method:** a drop is held still when the electric force balances its weight: **QV/d = mg**. Example: a drop of mass 2.0×10^{-15} kg is held between plates 1.2 cm apart by 490 V. $Q = mgd/V = 2.0 \times 10^{-15} \times 9.81 \times 0.012 \div 490 = 4.8 \times 10^{-19}$ C = **3e**. All charges come out as whole-number multiples of e, so charge is **quantised** (WPH14 Jan 2021, Jan 2025).
- The drop's radius comes from its terminal velocity with Stokes' law, $6\pi\eta rv = (4/3)\pi r^3 \rho g$ (WPH14 Jan 2021).
- A charged sphere touching a plate takes that plate's sign, is repelled to the other plate, swaps sign and returns: it carries charge across and a current flows (WPH14 Oct 2023).

SPEC 98

Potential in a radial field

WHAT THIS SPEC POINT ASKS

Use $V = Q/4\pi\varepsilon_0 r$ for the potential near a point charge.

$$V = Q / 4\pi\varepsilon_0 r$$

- V = potential (V), Q = the charge making the field (C), r = distance from its centre (m).
- **$V \propto 1/r$** (but $E \propto 1/r^2$). Triple the distance and the potential falls to a third.
- Sign follows Q: positive charges make positive potentials, negative charges negative potentials.
- **Electric potential energy** of a second charge q at distance r: $E_p = qV = Qq/4\pi\varepsilon_0 r$.

- **Several charges:** potentials are scalars, so just **add them with their signs**.

WORKED EXAMPLE: POTENTIAL AND WORK DONE

A +4.0 nC point charge. Potential at 0.30 m: $V = 8.99 \times 10^{9} \times 4.0 \times 10^{-9} \div 0.30 =$ **120 V**.
At 0.10 m: V = 360 V. Work done moving a +2.0 nC charge from 0.30 m to 0.10 m:
$W = q\Delta V = 2.0 \times 10^{-9} \times (360 - 120) =$ **4.8×10^{-7} J**

98 Closest approach of an alpha particle

An alpha particle fired straight at a nucleus slows down as it climbs the potential "hill". At the closest approach it is momentarily at rest, so **all its kinetic energy has become electrical potential energy**:

$$E_k = Q_\alpha Q_N / 4\pi\varepsilon_0 r_{min}$$

WORKED EXAMPLE: HOW CLOSE DOES A 5.0 MEV ALPHA GET TO A GOLD NUCLEUS?

$E_k = 5.0 \times 10^{6} \times 1.60 \times 10^{-19} = 8.0 \times 10^{-13}$ J. Charges: alpha = 2e, gold = 79e.
$r = (8.99 \times 10^{9} \times 2 \times 79 \times (1.60 \times 10^{-19})^2) \div 8.0 \times 10^{-13} =$ **4.5×10^{-14} m**
This is an **upper limit** for the radius of the gold nucleus: the alpha never touched it.

MARK-SCHEME LANGUAGE

- "Recognises Q is 2 (× unit charge)" for an alpha: a separate mark.
- Why the formula works for a nucleus: "the field is **radial**", "the nucleus can be treated as a **point charge**", "no other charges nearby", "distance measured from the **centre**".
- Two charges: "potential at each point is the **sum** of the potential due to X and the potential due to Y".
- Alpha not aimed head-on: it still has speed at its nearest point, so "it is not as close to the nucleus" and the maximum force is smaller.

Often asked as: potential or potential energy at a distance; work done moving a charge between radii; closest approach from an energy in MeV; p.d. between two points near two charges.

COMMON MISTAKE

Forgetting that gold is 79e and alpha is 2e (or using charge 1).
Neglecting the "M" in MeV.
Using r^2 (that is the field formula).
Adding potentials as vectors: they are scalars.

BEYOND THE SPEC, BUT EXAMINED

- **Hydrogen atom energy:** the potential energy of the electron is negative, $-e^2/4\pi\varepsilon_0 r = -4.3 \times 10^{-18}$ J at 5.3×10^{-11} m (WPH14 Jan 2026).
- **Maximum acceleration** at closest approach: F from Coulomb's law at r_{min}, then a = F/m (WPH14 Jan 2021).

SPEC 99 Field lines and equipotentials

WHAT THIS SPEC POINT ASKS

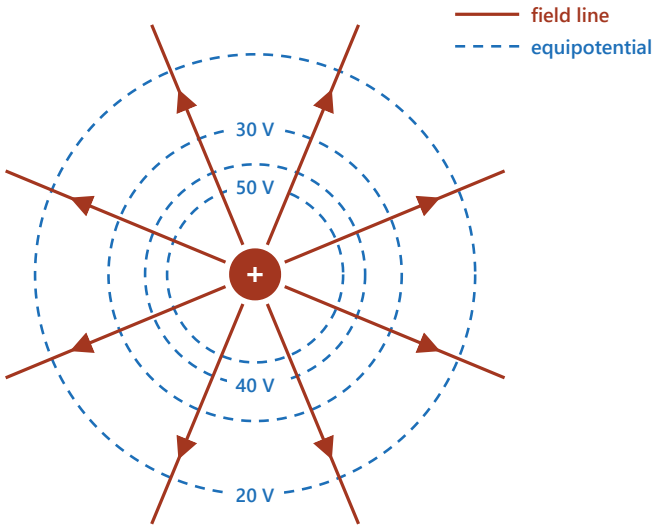
Draw and interpret field-line and equipotential diagrams for radial and uniform electric fields.

99 Rules for field lines

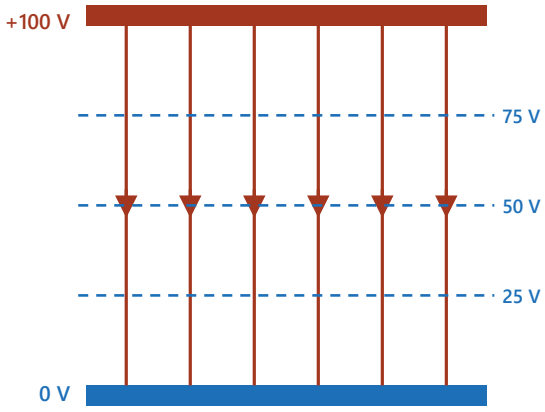
- A field line shows the direction of the force on a **positive** charge: draw an **arrow** on every line.
- Lines start on positive charges and end on negative charges (or go off to infinity).
- **Closer lines = stronger field**. Equally spaced parallel lines = **uniform** field.
- Lines **never cross** (the field can only have one direction at a point).
- Lines meet a conducting surface (a plate or sphere) at **90°**.

99 Rules for equipotentials

- An **equipotential** is a line (or surface in 3-D) joining points at the **same potential**.
- Equipotentials are always **perpendicular to field lines**.
- **No work** is done moving a charge along an equipotential ($\Delta V = 0$, so $Q\Delta V = 0$).
- Draw them at **equal steps in potential** (e.g. every 10 V). Where they are **closer** together the field is **stronger** ($E = -\Delta V/\Delta x$).
- The surface of a conductor is an equipotential.



Radial field: equipotentials spread out



Uniform field: equally spaced lines and equipotentials

Left: a positive point charge. Field lines are radial with arrows outwards; equipotentials (here every 10 V) are circles that get further apart. Right: parallel plates. Field lines are parallel, equally spaced and touch both plates; equipotentials are equally spaced lines parallel to the plates.

99 Radial field (point charge or charged sphere)

- Field lines: straight, **radial**, equally spaced in angle, arrows **out** of a positive charge and **in** to a negative one. They spread out, showing the field weakening.
- Equipotentials: **concentric circles**. Because $V \propto 1/r$, equal steps in V give circles that are **further apart** as r increases.

V	120 V	90 V	60 V	30 V
r (for V × r = 36 V m)	0.30 m	0.40 m	0.60 m	1.20 m
gap to next	0.10 m	0.20 m	0.60 m	

99 Uniform field (parallel plates)

- Field lines: **parallel, equally spaced**, straight, from the + plate to the − plate, touching both. (Ignore slight bulging at the edges.)
- Equipotentials: **parallel to the plates**, **equally spaced**. With 100 V across, the 50 V line is halfway.

99 Two charges (sketches you should recognise)

- **Opposite charges:** lines curve from + to −, crowded between them. The 0 V equipotential is the line halfway between equal and opposite charges.
- **Like charges:** lines from each charge bend away from the other; there is a **neutral point** (E = 0) between them. For unequal charges the neutral point is **nearer the smaller** charge.

MARK-SCHEME LANGUAGE

- Radial: "at least **four** radial straight lines", "**equally spaced**", "arrow(s) **outward**" (or inward for negative).
- Uniform: "at least three **equi-spaced** parallel lines", "**touching the plates**", "arrow". Curved lines at the edges are ignored.
- Equipotentials round a point charge: "circles centred on the charge" and "**spacing increasing** with distance".
- Two like charges: "neutral point equidistant", "lines diverging", "lines closer between the charges than outside".

Often asked as: draw the field round a charged sphere or between plates; add equipotentials; read the sign of charges from arrows; "is the field line the path the charge follows?"

COMMON MISTAKE

Drawing radial equipotentials **equally spaced**. They must spread out, because $V \propto 1/r$.
Missing arrows or arrows the wrong way. Use a ruler.
Lines that stop short of the plates, or that cross.

EXAM TIP

A field line is **not** the path of a charge in general. A charge released at rest on a **straight** field line does move along it, but on a curved line its velocity cannot follow the changing direction of the force.

SPEC 100 Capacitance: C = Q/V

WHAT THIS SPEC POINT ASKS

Capacitance is defined as C = Q/V; use this equation.

100 What a capacitor does

- A capacitor is two conducting plates separated by an insulator (the **dielectric**).
- When connected to a supply, electrons are pushed onto one plate and pulled off the other. One plate gets charge **+Q**, the other **−Q**. The **total** charge on the capacitor is zero; it **separates** charge.
- "Charge stored Q" means the charge on **one** plate.
- The p.d. across the plates rises as charge builds up. Q is **proportional** to V.

$$C = Q / V$$

- C = capacitance (farad, F), Q = charge on one plate (C), V = p.d. across the capacitor (V). 1 F = 1 C V^{-1}.
- A farad is huge. Typical values: μF (10^{-6} F), nF (10^{-9} F), pF (10^{-12} F).
- On a graph of **Q against V**, the gradient is C. On a graph of **V against Q**, the gradient is **1/C**.

WORKED EXAMPLE: CHARGE STORED

A 470 μF capacitor is charged to 9.0 V.
$Q = CV = 470 \times 10^{-6} \times 9.0 =$ **4.2×10^{-3} C**

MARK-SCHEME LANGUAGE

- Capacitors "separate charge"; "the total charge on a capacitor is zero".
- Sharing: when a charged capacitor is connected to an uncharged one, "the charge stored in each is **less** than the original, because the **p.d. is shared**".

Often asked as: Q = CV after finding C from a time constant; initial current V/R; MCQ on the gradient of a V–Q or Q–V graph.

COMMON MISTAKE

Mixing up C (capacitance, in F) with C (the coulomb, unit of charge). Write units carefully.
Answering a sharing question in terms of "capacitance changes" when it is the p.d. that is shared.

SPEC 101

Energy stored by a capacitor

WHAT THIS SPEC POINT ASKS

Use $W = \frac{1}{2}QV$; derive it from the area under a p.d.–charge graph; derive and use $W = \frac{1}{2}CV^2$ and $W = \frac{1}{2}Q^2/C$.

101 Derivation from the V–Q graph

- While charging, the p.d. across the capacitor rises in proportion to the charge: the graph of V against Q is a **straight line through the origin** (gradient 1/C).
- To add a small charge ΔQ at p.d. V, work done = VΔQ: a thin strip under the graph.
- Adding all the strips: **energy stored = area under the V–Q graph** = area of a triangle = ½ × Q × V.

$$W = \tfrac{1}{2}QV = \tfrac{1}{2}CV^2 = \tfrac{1}{2}Q^2/C$$

- W = energy stored (J), Q = charge (C), V = p.d. (V), C = capacitance (F).
- The other forms come from substituting Q = CV (gives $\frac{1}{2}CV^2$) or V = Q/C (gives $\frac{1}{2}Q^2/C$).
- Energy ∝ V^2: **double the voltage, four times the energy**.

101 Why it is ½QV and not QV

The supply does work **QV** pushing charge Q through p.d. V. Only **half** ends up stored in the capacitor; the other half is dissipated as heat in the resistance of the circuit while it charges.

WORKED EXAMPLE: ENERGY STORED AND RELEASED

A 2200 μF capacitor is charged to 12 V.
$W = \frac{1}{2}CV^2 = 0.5 \times 2200 \times 10^{-6} \times 12^2 =$ **0.16 J**
It discharges to 6.0 V. Energy released = $\frac{1}{2}C(V_1^2 - V_2^2) = 0.5 \times 2.2 \times 10^{-3} \times (144 - 36) =$ **0.12 J**
The p.d. halved but **three-quarters** of the energy was released.
With a +20% tolerance on C, the maximum energy at 12 V is 1.2 × 0.158 = 0.19 J.

MARK-SCHEME LANGUAGE

- "Use of $W = \frac{1}{2}CV^2$" (or $\frac{1}{2}QV$ or $\frac{1}{2}Q^2/C$).
- "Use of tolerance as +20%" is a separate mark: take the extreme value in the direction the question needs.
- Comparisons need a conclusion: "184 J > 150 J, so enough energy".

Often asked as: energy stored; energy released between two voltages; tolerance questions; comparing with another energy store (battery, spring); MCQ on how W changes when V changes.

COMMON MISTAKE

Forgetting to square V.
Leaving V in mV or C in μF.
Taking energy released as $\frac{1}{2}C(V_1 - V_2)^2$. It is $\frac{1}{2}C(V_1^2 - V_2^2)$: the **difference of two energies**.

BEYOND THE SPEC, BUT EXAMINED

The battery does work QV while the capacitor stores only $\frac{1}{2}QV$ (WPH14 Oct 2021 MCQ).

SPEC 102

Charging and discharging curves; the time constant RC

WHAT THIS SPEC POINT ASKS

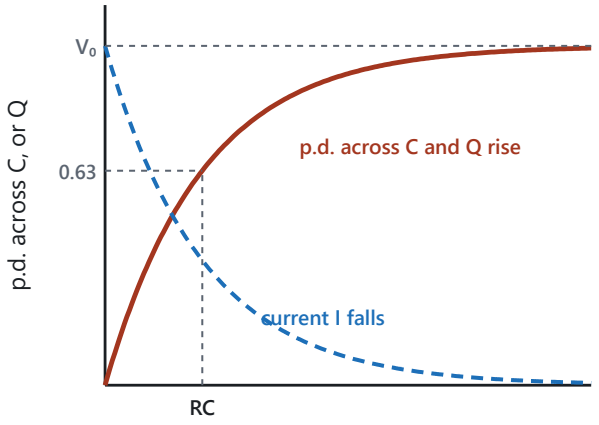
Draw and interpret charge and discharge curves for RC circuits, and understand the significance of the time constant RC.

102 Charging through a resistor (from a supply V_0)

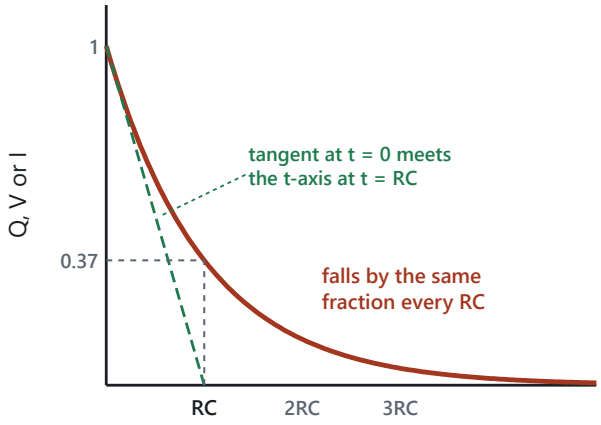
1. At the start the capacitor has no charge, so $V_C = 0$, all of V_0 is across R, and the current is **maximum**: $I_0 = V_0/R$.
2. Charge builds up on the plates, so the **p.d. across C rises**.
3. Since $V_R + V_C = V_0$, the **p.d. across R falls**, so the **current falls** ($I = V_R/R$).
4. Charge therefore arrives more slowly: Q and V_C rise more and more slowly.
5. Finally $V_C = V_0$ and **$I = 0$**.

102 Discharging through a resistor

- At the start $V = V_0$ and $I_0 = V_0/R$. As charge flows off, V falls, so I falls, so the charge flows off more slowly.
- Q, V and I **all** decay **exponentially**: each falls by the same fraction in each equal time interval.



Charging through R from a supply V_0



Discharging through R (fraction of start value)

Left: charging. The p.d. across C (and Q) rise to V_0, reaching 63% after one time constant; the current (dashed, on its own scale) falls from I_0. Right: discharging. Q, V and I all fall to 37% of their start value after RC; the tangent at t = 0 cuts the time axis at t = RC.

102 The time constant τ = RC

$$\tau = RC$$

- τ = time constant (s), R = resistance (Ω), C = capacitance (F). $\Omega \times F = (V\,A^{-1})(C\,V^{-1}) = C\,A^{-1} = s$.
- **Discharging:** after one time constant Q, V and I have fallen to **1/e ≈ 37%** of their initial values.
- **Charging:** after one time constant Q and V_C have risen to **63%** of their final values.
- After about **5RC** the capacitor is effectively fully charged or discharged (less than 1% left).
- Time to halve: $t_{½} = RC \ln 2 \approx$ **0.69RC**.
- Bigger R or bigger C → bigger τ → **slower** charging and discharging.

WORKED EXAMPLE: TIME CONSTANT

R = 47 kΩ, C = 220 μF. $\tau = RC = 47 \times 10^3 \times 220 \times 10^{-6} =$ **10 s**.
Time to fall to half: $t_{½} = 0.69 \times 10.3 =$ **7.2 s**.

MARK-SCHEME LANGUAGE

Six-mark charging explanation (indicative content):

- "There is a **current** in the circuit; the **charge on the capacitor increases**."
- "The **p.d. across the capacitor increases**, so the **p.d. across the resistor decreases**."
- "So the **current decreases** (I = V/R) and the rate of charging falls."
- "Finally the p.d. across C equals the supply p.d. and the current is zero."

Sketches: "exponential curve from the origin levelling at V_0", "passes about 2/3 of the maximum at t = RC", "levels off after 4–5 time constants"; discharge "starts on the y-axis" and "does not cross or crash into the t-axis". Label the axes with quantities and units.

Often asked as: sketch a charging or discharging graph of V or I; explain the shape; effect of a changed resistance on the time constant; where $V_R = V_C$ (each half the supply).

COMMON MISTAKE

Explaining with "charges moving" but never mentioning **current** and **p.d.**: no marks.
Using 37% on a **charging** curve (it is 63% for charging).
Saying the effect of a change without going through the time constant or current first.
Curves that hit the time axis.

BEYOND THE SPEC, BUT EXAMINED

- Components whose resistance changes with p.d. (an LED or diode in the discharge path): as V falls, R rises, so the time constant **increases** (WPH14 Jan 2020).
- A capacitor as a **backup supply**: its p.d. "remains high enough to operate the device for a short time" (WPH14 Oct 2021).

SPEC 103

Core Practical 11: capacitor charge and discharge

WHAT THIS SPEC POINT ASKS

Use an oscilloscope or data logger to display and analyse the p.d. across a capacitor as it charges and discharges through a resistor.

CORE PRACTICAL 11: METHOD

Circuit: d.c. supply (e.g. 6.0 V), two-way switch, resistor R (e.g. 100 kΩ) and capacitor C (e.g. 220–470 μF) in series. A **voltmeter, data logger or oscilloscope** goes **in parallel with the capacitor**. In one switch position the supply charges C through R; in the other, C discharges through R alone.

1. Set the data logger to record V against time at a suitable rate (several readings per second for τ of tens of seconds).
2. Switch to charge. Record V until it stops changing (about 5RC).
3. Switch to discharge and record V as it falls.
4. Repeat for other values of R (or C).

Oscilloscope version: replace the supply and switch with a **square-wave signal generator**, so the capacitor charges and discharges repeatedly. Adjust the time-base so one charge–discharge cycle fills the screen, then read V and t from the grid. The period of the square wave should be much longer than RC (about 10RC).

Safety: connect an **electrolytic** capacitor the right way round (+ to +) and do not exceed its working voltage, or it can overheat and burst.

103 Analysis

- From $V = V_0e^{-t/RC}$, taking natural logs: **$\ln V = \ln V_0 - t/RC$**.
- Plot **ln V** (y) against **t** (x): a straight line with **gradient $-1/RC$** and intercept $\ln V_0$.
- So $RC = -1/\text{gradient}$, and $C = -1/(R \times \text{gradient})$.
- Quick checks: the time to fall to $0.37V_0$ (discharge) or rise to $0.63V_0$ (charge) is RC.

WORKED EXAMPLE: CAPACITANCE FROM A LN V GRAPH

R = 100 kΩ. The best-fit line on a graph of ln V (V in volts) against t (in s) passes through (0, 1.79) and (40, 0.46).
Gradient = (0.46 − 1.79) ÷ 40 = −0.0333 s^{-1}
RC = 1 ÷ 0.0333 = 30.1 s, so C = 30.1 ÷ 1.00 × 10^5 = **3.0 × 10^{-4} F** (300 μF)

103 Errors and improvements

- Capacitors have large **tolerances** (often ±10% or ±20%), so compare your C with the nominal value using the tolerance.
- A voltmeter with too low a resistance gives a second discharge path and makes the discharge look faster. A data logger or oscilloscope has a very high resistance.
- Reading by eye with a stopwatch is limited by reaction time: choose RC of at least tens of seconds, or use a data logger.
- Use the best-fit line over many points rather than one pair of readings.

MARK-SCHEME LANGUAGE

- Circuit: "battery, resistor and capacitor in **series**", "voltmeter / data logger **in parallel with the capacitor**", "switch arranged to give a **discharge path** through the resistor".
- Why a given circuit is wrong: "when charging the voltmeter is not across C" or "when discharging the resistor is not in the circuit".
- Tolerance: "6640 Ω is within 5% of 6800 Ω" with the calculation shown.

Often asked as: draw the circuit; read V and t off a data-logger or oscilloscope trace; find R or C from it; check a component is within tolerance.

BEYOND THE SPEC, BUT EXAMINED

A repeating charge–discharge trace: energy released per discharge = ½C(V_1^2 − V_2^2) (WPH14 Jun 2025).

SPEC 104 Exponential discharge equations

WHAT THIS SPEC POINT ASKS

Use Q = Q_0e^(−t/RC), derive and use I = I_0e^(−t/RC) and V = V_0e^(−t/RC), and the matching log equations.

$$Q = Q_0 e^{-t/RC} \quad I = I_0 e^{-t/RC} \quad V = V_0 e^{-t/RC}$$

- Q_0, I_0, V_0 = values at t = 0; t = time since discharge began (s); RC = time constant (s).
- **Deriving the V form:** V = Q/C, and C is constant, so V = $Q_0 e^{-t/RC}/C = V_0 e^{-t/RC}$.
- **Deriving the I form:** during discharge the p.d. across R equals V, so I = V/R = $(V_0/R)e^{-t/RC} = I_0 e^{-t/RC}$.

104 Log forms

$$\ln Q = \ln Q_0 - t/RC \quad \ln I = \ln I_0 - t/RC \quad \ln V = \ln V_0 - t/RC$$

- Each has the form y = c + mx: plot ln Q (or ln I, ln V) against t to get a straight line of **gradient −1/RC**.
- Rearranged to find a time: **$t = RC \ln(V_0/V)$**.

104 What fraction is left?

Time	1RC	2RC	3RC	4RC	5RC
Fraction left (discharge), e^{-n}	0.37	0.14	0.050	0.018	0.0067
Fraction reached (charging), $1 - e^{-n}$	0.63	0.86	0.95	0.98	0.99

WORKED EXAMPLE: CURRENT AND TIME DURING DISCHARGE

C = 100 μF charged to 12 V discharges through R = 2.0 kΩ. RC = $2.0 \times 10^3 \times 100 \times 10^{-6}$ = 0.20 s; I_0 = 12 ÷ 2000 = 6.0 mA.

Current after 0.30 s: $I = 6.0 \times e^{-0.30/0.20} = 6.0 \times 0.223 =$ **1.3 mA**

Time for V to fall to 3.0 V: $t = RC \ln(V_0/V) = 0.20 \times \ln(12/3.0) = 0.20 \times 1.386 =$ **0.28 s**

104 Charging equations

- During charging the p.d. across the **resistor** decays: $V_R = V_0 e^{-t/RC}$, and so does the current.
- Because $V_C = V_0 - V_R$:

$$V_C = V_0(1 - e^{-t/RC}) \quad Q = Q_0(1 - e^{-t/RC})$$

- Example: the same circuit charged from 12 V reaches 9.0 V when $e^{-t/0.20} = 0.25$, so $t = 0.20 \times \ln 4 = 0.28$ s.

104 Ways to find RC from a graph (all credited)

1. **Time-constant route:** read the time when $V = 0.37V_0$ (discharge) or $0.63V_0$ (charge).
2. **Tangent route:** draw the tangent at t = 0; it meets the t-axis (or the final-value line) at t = RC.
3. **Substitution route:** take a pair of (t, V) values and solve $V = V_0 e^{-t/RC}$ using logs.
4. **Half-life route:** read $t_{½}$, then $RC = t_{½}/\ln 2$.
5. **Log-graph route:** gradient of ln V against t = −1/RC.

MARK-SCHEME LANGUAGE

- "Use of $V = V_0 e^{-t/RC}$ **Or** use of $\ln V = \ln V_0 - t/RC$", "use of time constant = RC".
- "Takes corresponding pairs of values of V and t from the graph."
- Choosing a standard capacitor: pick the value that keeps the time **safe** (e.g. calculated 0.56 F → choose 0.58 F so the time is not too short).
- Answers read from graphs are marked within a range.

Often asked as: C or R from a graph; time to reach a given p.d. or current; derive the charging equation; percentage left after n time constants; tolerance checks.

COMMON MISTAKE

Putting values from a **charging** graph straight into the **decay** equation. Use $V_0 - V_C$ (the resistor's p.d.) or the $(1 - e^{-t/RC})$ form.

Taking V_0 as the highest reading (5.8 V) instead of the supply p.d. (6.0 V).

Finding "the time for V to fall by a factor of e" when RC itself was asked for, and vice versa: read the question.

Using "fully charged after 5RC" as a calculation method.

No unit on the final answer.

SPEC 105 Magnetic flux density, flux and flux linkage

WHAT THIS SPEC POINT ASKS

Understand and use magnetic flux density B, magnetic flux φ and flux linkage Nφ.

105 Magnetic flux density B

- B measures the **strength** of a magnetic field: the force per unit current per unit length on a wire at **right angles** to the field (from F = BIl).
- Unit: the **tesla** (T). 1 T = 1 N A^{-1} m^{-1}. Earth's field is about 50 μT; a strong lab magnet about 0.1–1 T.
- B can be thought of as the **density of field lines**: the flux per unit area.

105 Magnetic flux φ

$$\varphi = BA \quad (\text{B perpendicular to the area}) \qquad \varphi = BA\cos\theta \quad (\text{in general})$$

- φ = flux (weber, Wb; 1 Wb = 1 T m^2), A = area (m^2), θ = angle between B and the **normal** to the area.
- Plane of the coil perpendicular to B: maximum flux BA. Plane of the coil parallel to B: zero flux.

105 Flux linkage Nφ

$$\text{flux linkage} = N\varphi = BAN$$

- N = number of turns. Unit: Wb turns (Wb is accepted).
- Induced e.m.f. depends on how fast the **flux linkage** changes (110).

WORKED EXAMPLE: FLUX LINKAGE OF A TILTED COIL

A 200-turn coil of radius 1.5 cm is in a field of 0.060 T. The field makes 30° with the normal to the coil.

$A = \pi r^2 = \pi \times 0.015^2 = 7.07 \times 10^{-4}$ m^2

$\varphi = BA\cos\theta = 0.060 \times 7.07 \times 10^{-4} \times \cos 30° = 3.67 \times 10^{-5}$ Wb

$N\varphi = 200 \times 3.67 \times 10^{-5} =$ **7.3×10^{-3} Wb turns**

MARK-SCHEME LANGUAGE

- "Calculates area of coil", "applies flux = flux density × area", "flux linkage = Nφ". Units: "Wb, weber turns or T m^2".
- Component of a field through a tilted surface: e.g. "$B_{vertical}$ = 55 μT × sin 65°".

Often asked as: φ or Nφ for a coil; unit of flux linkage; MCQ on flux linkage after turning a coil through 180° (it becomes −Nφ).

COMMON MISTAKE

Using the diameter in πr^2.
Multiplying by N twice (once in φ, again for flux linkage).
Taking θ as the angle to the **plane** of the coil but still using cos. Check whether θ is measured to the normal or to the surface.

BEYOND THE SPEC, BUT EXAMINED

The Earth's field has horizontal and vertical components. Over the equator it is horizontal, so a level aircraft wing cuts no flux; elsewhere use the vertical component for a horizontal wing (WPH14 Oct 2025).

SPEC 106 Force on a charge moving in a magnetic field

WHAT THIS SPEC POINT ASKS

Use F = Bqv sin θ and Fleming's left-hand rule for charged particles moving in a magnetic field.

$$F = Bqv \sin \theta$$

- F = force (N), B = flux density (T), q = charge (C), v = speed (m s^{-1}), θ = angle between v and B.
- Maximum force (F = Bqv) when v is **perpendicular** to B; **zero** force when the particle moves **parallel** to B.

106 Direction: Fleming's left-hand rule

- **F**irst finger = **F**ield (N to S), se**C**ond finger = **C**urrent, thu**M**b = **M**otion (force). The three are at right angles.
- The "current" finger points along the **conventional current**: the direction a **positive** charge moves. For an **electron**, point the second finger **opposite** to its velocity.
- Notation: **×** = field into the page, **•** = field out of the page.

106 Consequences

- The force is always **perpendicular to the velocity**, so it does **no work**: the **speed and kinetic energy stay constant**. Only the direction changes.
- In a uniform field at right angles, the force is a constant-size centripetal force, so the particle moves in a **circle** (see 115).

WORKED EXAMPLE: FORCE ON A PROTON

A proton moves at 3.0×10^6 m s^{-1} at right angles to a 0.25 T field.
F = Bqv = $0.25 \times 1.60 \times 10^{-19} \times 3.0 \times 10^6$ = **1.2×10^{-13} N**
At 40° to the field: F = $1.2 \times 10^{-13} \times \sin 40°$ = **7.7×10^{-14} N**

MARK-SCHEME LANGUAGE

- "Use of F = Bqv sin θ" (sin θ = 1 when perpendicular).
- Direction marks: "applying FLHR, field **into** the page", often dependent on a correct direction of motion.
- Velocity selector: "magnetic force = electric force", BQv = EQ, so **v = E/B**.

Often asked as: calculate F, v or B; field direction from the way a track curves; velocity selector and e/m questions.

COMMON MISTAKE

Forgetting to reverse the rule for negative particles.
Using charge = 1, or B in mT without converting.
Mixing up F = Bqv (a moving charge) with F = BIl (a wire).

BEYOND THE SPEC, BUT EXAMINED

- **Velocity selector:** crossed E and B fields; only particles with v = E/B (= V/Bd for plates) pass undeflected. Example: E = 2.0 × 10^4 V m^{-1} and B = 5.0 mT select v = 4.0 × 10^6 m s^{-1} (WPH14 Jan 2023, Oct 2022).
- **Measuring e/m:** combine the selected speed with the radius in the magnetic field alone: e/m = v/Br (WPH14 Jan 2023, Oct 2023).

SPEC 107

Force on a current-carrying conductor

WHAT THIS SPEC POINT ASKS

Use F = BIl sin θ and Fleming's left-hand rule for a current-carrying conductor in a magnetic field.

$$F = BIl \sin \theta$$

- F = force (N), B = flux density (T), I = current (A), l = length of wire **in the field** (m), θ = angle between the wire and B.
- Maximum when the wire is **perpendicular** to the field; zero when **parallel**.
- Link to 106: a current is moving charge. Charge Q passing along length l in time t gives I = Q/t and v = l/t, so BIl = B(Q/t)(vt) = BQv.
- Direction by **Fleming's left-hand rule**, with the second finger along the **conventional current**.

WORKED EXAMPLE: WIRE ON A BALANCE

A 5.0 cm length of wire carrying 3.5 A sits at right angles to a 0.12 T field.
F = BIl = 0.12 × 3.5 × 0.050 = **0.021 N**
On a top-pan balance under the magnet this shows as a change in reading of 0.021 ÷ 9.81 = 2.1 × 10^{-3} kg = **2.1 g** (Newton's third law: the wire pushes back on the magnet).

WORKED EXAMPLE: CABLE IN THE EARTH'S FIELD

A 30 m cable carries 200 A at 60° to the Earth's field of 50 μT.
F = BIl sin θ = 50 × 10^{-6} × 200 × 30 × sin 60° = **0.26 N**

MARK-SCHEME LANGUAGE

- "Component of B calculated using cos 20° **or** sin 70°": use the component perpendicular to the wire.
- Coil in a field: "by FLHR, force **up** on one side, **down** on the other", forming a couple.
- Balance questions: "magnetic force = weight", or the balance reading change = F/g.

Often asked as: balance or rod experiments to find B; a cable in the Earth's field; forces on the sides of a coil.

COMMON MISTAKE

Mixing up sin and cos when finding the field component.
Explaining a motor-effect question with induction and Lenz's law.
Giving the force on one side of a coil only, when both are asked for.

BEYOND THE SPEC, BUT EXAMINED

- **Couple on a coil:** each side of an N-turn coil feels NBIl; the moment = force × perpendicular distance between the sides (WPH14 Oct 2024).
- **Rod on rails:** F = BIl accelerates the rod; with suvat, B = ma/Il (WPH14 Oct 2025).

SPEC 108

E.m.f. from relative motion of a coil and a magnet

WHAT THIS SPEC POINT ASKS

Understand the factors that affect the e.m.f. induced in a coil when a coil and a permanent magnet move relative to each other.

108 What happens

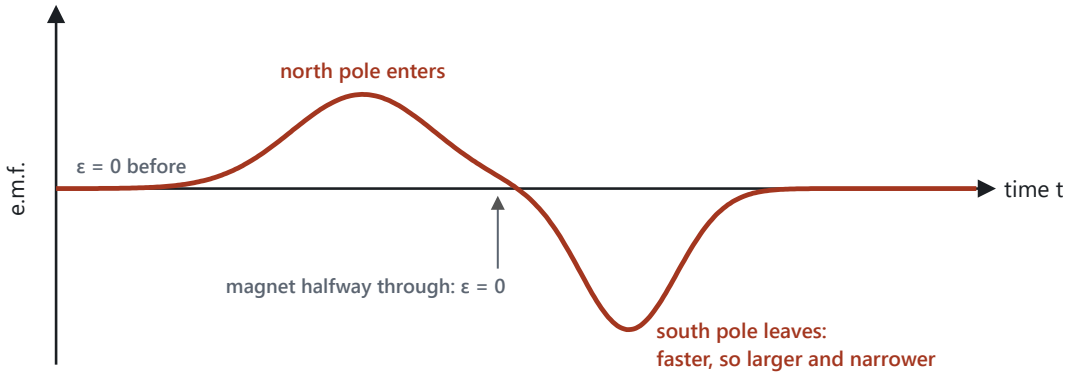
- Move a magnet into a coil: the flux linking the coil changes, so an **e.m.f. is induced** and, if the circuit is complete, a current flows.
- Hold the magnet still inside the coil: no change in flux linkage, **no e.m.f.**
- Pull it out: e.m.f. in the **opposite** direction. Reversing the pole also reverses it.
- It is the **relative** motion that matters: moving the coil over a still magnet works too.
- Equivalent picture: the wires of the coil **cut lines of flux**.

108 Factors that increase the e.m.f.

All of these increase the **rate of change of flux linkage**:

- **faster** relative motion
- **stronger** magnet (larger B)
- **more turns** on the coil (larger N)
- larger coil area, or a soft-iron core inside the coil

108 A magnet falling through a coil



As the north pole enters, flux linkage increases: positive pulse. Halfway through the flux linkage is momentarily constant: ε = 0. As the south pole leaves, flux linkage decreases: the e.m.f. reverses. The magnet is moving faster by then, so the second pulse is taller and narrower. The two areas are equal, because the flux linkage starts and ends at zero.

108 A straight wire moving through a field

A wire of length l moving at speed v at right angles to B sweeps out area lv each second, so the flux cut per second is Blv:

$$\varepsilon = Blv$$

108 A rotating coil (a generator)

As a coil rotates steadily in a uniform field, its flux linkage rises and falls sinusoidally, so it produces an **alternating** e.m.f. Faster rotation gives a larger peak e.m.f. **and** a higher frequency (see the graphs in 110).

MARK-SCHEME LANGUAGE

Magnet falling through a coil (six-mark style):

- "**Change of flux linkage** with the coil" or "wires **cut lines of flux**" → "e.m.f. induced".
- "The greater the **rate of change** of flux linkage, the larger the e.m.f."
- "After the magnet's midpoint passes, flux linkage **decreases**, so the e.m.f. is **negative**."
- "The magnet speeds up, so the second peak is **larger** and **narrower**."
- "Zero e.m.f. before the magnet enters and when its midpoint is in the coil."
- "Magnet cuts lines" is ignored: it is the **wires** that cut the flux.

Often asked as: explain the e.m.f.–time trace; magnets passing a card reader; aircraft wings in the Earth's field; magnet falling through a copper tube.

COMMON MISTAKE

Writing "change in flux" without "linkage **with the coil**".
Bringing in Lenz's law when the question only asks about the size of the e.m.f.
Missing that it is the **change in direction** of the e.m.f. that carries the information (card readers).

BEYOND THE SPEC, BUT EXAMINED

- **Magnet falling through a metal tube:** currents induced in the tube create a field that opposes the motion (Lenz), so the magnet falls slowly; a tube with a slit falls faster because the current cannot circulate (WPH14 Oct 2022, 6-mark).
- **Order-of-magnitude checks** with ε = Blv for aircraft wings (WPH14 Jan 2024).

SPEC 109 E.m.f. from a changing current in a linked coil

WHAT THIS SPEC POINT ASKS

Understand the factors that affect the e.m.f. induced in a coil when the current changes in another coil linked with it.

109 The chain of cause and effect

1. A current in the **primary** coil produces a **magnetic field**.
2. When that current **changes**, the field changes.
3. So the **flux linkage with the secondary** coil changes.
4. So an **e.m.f. is induced** in the secondary (and a current, if its circuit is complete).

109 What gives an e.m.f., and how big

- **Steady d.c.:** no change, **no e.m.f.**
- **Switching on:** current rises, brief e.m.f. one way. **Switching off:** current falls, brief e.m.f. the **other** way (often bigger, because the current collapses faster).
- **a.c.:** the current is always changing, so there is an alternating e.m.f. of the **same frequency**.

- The e.m.f. is larger with: a **faster rate of change** of current (larger current or higher frequency), **more turns** on the secondary (and primary), a **soft-iron core** linking both coils, and coils **closer** together and better aligned (more of the flux links the secondary).

109 Transformers (background)

A transformer is two coils on a shared iron core. An alternating current in the primary gives an alternating flux in the core, which links the secondary. For an ideal transformer, $V_s/V_p = N_s/N_p$ (not required by the spec, but it follows from Faraday's law).

MARK-SCHEME LANGUAGE

Wording that earns the marks:

- "current in the primary produces a magnetic field"
- "when the switch is opened the **current changes**, so there is a **change in flux linkage** in the secondary coil"
- "an e.m.f. is induced across the secondary".

Wireless charging, not all flux linking: "no iron core" or "coils too far apart".

Often asked as: six-mark chains (wireless charger, induction heating, plasma loop); switch-on/switch-off pulses; MCQ on how to increase the induced e.m.f.

COMMON MISTAKE

Saying only "a change in flux" without naming the **changing current** and the **flux linkage with the secondary**.
Using "opposes the motion" for Lenz's law when nothing is moving.
Putting the induced currents in the wrong object (in an induction hob, they flow in the **pan**, not the coil).

BEYOND THE SPEC, BUT EXAMINED

Induction heating and eddy currents. An alternating current in a coil gives an alternating field, so the flux linkage with a metal pan (or kettle base) changes, inducing an e.m.f. and **circulating (eddy) currents** in the metal, which heat it ($P = I^2R$). Only conductors heat up, because currents can only circulate in a conductor (WPH14 Jan 2024, 6-mark).

SPEC 110 Faraday's and Lenz's laws

WHAT THIS SPEC POINT ASKS

Use Faraday's law to find the size of an induced e.m.f., and the combined Faraday–Lenz equation $\varepsilon = -d(N\varphi)/dt$.

110 The two laws

- **Faraday's law:** the induced e.m.f. is equal to the **rate of change of flux linkage**.
- **Lenz's law:** the induced e.m.f. (and current) acts in a direction to **oppose the change that produces it**.

$$\varepsilon = -d(N\varphi)/dt \qquad \text{average e.m.f.: } \varepsilon = \Delta(N\varphi)/\Delta t$$

- ε = induced e.m.f. (V), $N\varphi$ = flux linkage (Wb turns), t = time (s). The minus sign is Lenz's law; drop it when you only want the size.

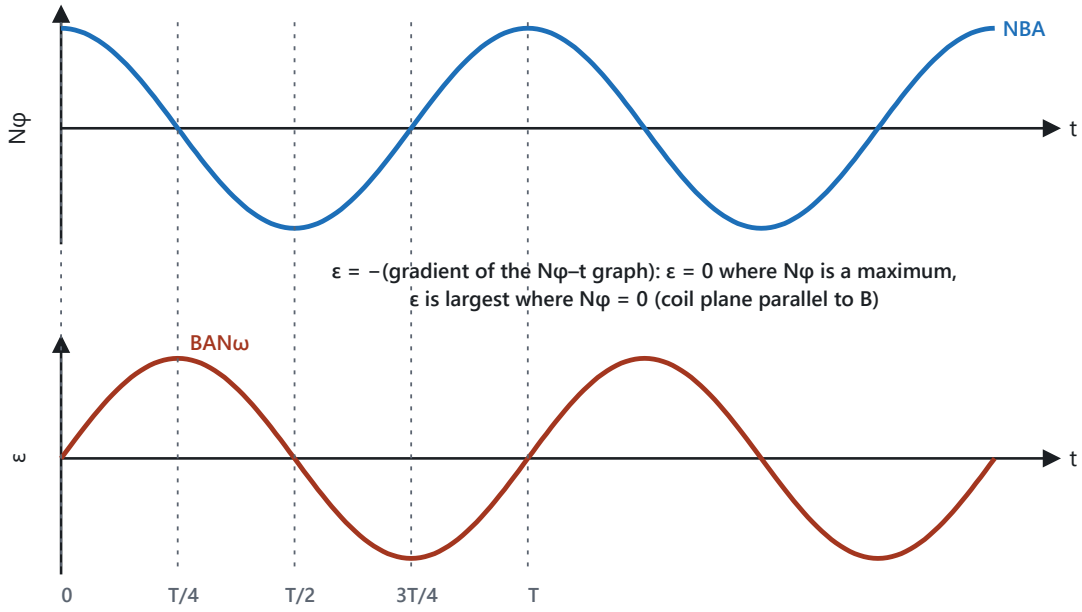
- On a graph of **Nφ against t**, the e.m.f. is the **gradient** (with a sign change). For the **maximum** e.m.f., draw a tangent at the **steepest** point.

110 Why Lenz's law must be true

If the induced current helped the change, it would speed the magnet up, giving more current, and so on: energy from nothing. Opposing the change means you must **do work** to move the magnet, and that work becomes the electrical energy. Lenz's law is **conservation of energy**.

Example: push a north pole into a coil. The induced current makes that end of the coil a **north pole**, which repels the magnet. Pull it out and that end becomes a south pole, attracting the magnet back.

110 A rotating coil



A coil of N turns and area A rotating at ω in field B. Flux linkage Nφ = BAN cos ωt; e.m.f. ε = BANω sin ωt. The e.m.f. is zero where the flux linkage is largest (coil face-on to the field) and largest where the flux linkage is zero (coil plane parallel to B, sides cutting flux fastest).

WORKED EXAMPLE: AVERAGE E.M.F. WHEN A COIL IS TURNED

A 150-turn coil of area 2.0×10^{-3} m^2 is face-on to a 0.050 T field. It is turned so its plane is parallel to the field in 0.12 s.

$\Delta(N\varphi) = NBA - 0 = 150 \times 0.050 \times 2.0 \times 10^{-3} = 0.015$ Wb turns

$\varepsilon = \Delta(N\varphi)/\Delta t = 0.015 \div 0.12 =$ **0.13 V**

If it rotates steadily at 50 revolutions per second: peak $\varepsilon = BAN\omega = 0.050 \times 2.0 \times 10^{-3} \times 150 \times (2\pi \times 50)$ = **4.7 V**

WORKED EXAMPLE: E.M.F. ACROSS AN AIRCRAFT WING

Wingspan 40 m, speed 250 m s^{-1}, vertical component of the Earth's field 40 μT.

In 1 s the wing sweeps area $40 \times 250 = 1.0 \times 10^4$ m^2, so flux cut per second = BA/t:

$\varepsilon = Blv = 40 \times 10^{-6} \times 40 \times 250 =$ **0.40 V**

110 Method for e.m.f. questions

1. Find the flux linkage: Nφ = NBA (area of the coil, or area swept).
2. Find the time it changes over, or the steepest gradient of a graph.
3. $\varepsilon = \Delta(N\varphi)/\Delta t$ (or Blv, or BANω for the peak of a rotating coil).

4. If asked: $I = \varepsilon/R$, $P = \varepsilon I$ or I^2R.
5. Compare with any value in the question and state a conclusion.

MARK-SCHEME LANGUAGE

- "Use of $\varepsilon = N\,d\varphi/dt$" (no minus sign needed for a magnitude); "evidence of attempt to find the **maximum gradient**".
- Lenz's law: "the induced e.m.f. (or current) is in a direction to **oppose the change** that produces it".
- Eddy-current braking chain: "change in flux linkage in the fin (or the fin cuts the field)" → "e.m.f. induced" → "(eddy) current flows" → "current in a magnetic field experiences a force" → "force opposes the motion (Lenz)" or "energy dissipated comes from the kinetic energy".
- Thicker plate: "lower resistance → larger current → larger braking force".

Often asked as: maximum e.m.f. from a graph; motional e.m.f. "show that" ($\varepsilon = Bdv$); peak e.m.f. of a rotating coil; power in an induced current; magnetic braking; direction of induced current or force.

COMMON MISTAKE

Using the wrong time for the change, or the mean e.m.f. when the peak is asked for.
Inventing factors (×2 "for two poles").
"Lenz's law: opposes the motion" when nothing moves; say it opposes the **change in flux linkage**.
Stating Lenz's law but not applying it to the situation in the question.

BEYOND THE SPEC, BUT EXAMINED

- **Eddy currents and magnetic braking.** A metal disc, fin or plate moving through a field has e.m.f.s induced in it; currents circulate, and the force on them opposes the motion. The kinetic energy becomes internal energy (heating). Used in train brakes and to damp balances. Asked in 7 of the 19 papers (WPH14 Oct 2021, Jun 2022).
- **Terminal speed of a rod on rails:** it stops accelerating when its induced e.m.f. equals the supply e.m.f. (WPH14 Oct 2025).
- **Resistivity inside an induction question:** $R = \rho l/A$, then $I = \varepsilon/R$ and $P = \varepsilon I$ (WPH14 Jan 2021).

Topic 3 – Nuclear and Particle Physics

SPEC 111 Nucleon number and proton number

WHAT THIS SPEC POINT ASKS

Understand what nucleon number (mass number) and proton number (atomic number) mean.

111 Nuclide notation

A nucleus of element X is written ${}^{A}_{Z}X$, for example ${}^{238}_{92}U$.

Symbol	Name	Meaning
A (top)	**nucleon number** (mass number)	total number of **protons + neutrons**
Z (bottom)	**proton number** (atomic number)	number of **protons**; fixes which element it is
N = A − Z	neutron number	number of neutrons

- Uranium-238: 92 protons, 238 − 92 = **146 neutrons**. A neutral atom also has 92 electrons.
- **Isotopes** have the same Z but different A (same number of protons, different numbers of neutrons), e.g. ${}^{1}_{1}H$, ${}^{2}_{1}H$, ${}^{3}_{1}H$.
- An **ion** has gained or lost electrons; its nucleus, and so A and Z, are unchanged.
- Particle symbols in this notation: proton ${}^{1}_{1}p$, neutron ${}^{1}_{0}n$, electron ${}^{0}_{-1}e$, positron ${}^{0}_{+1}e$, alpha ${}^{4}_{2}He$.

111 Balancing nuclear equations

Both totals must match on each side: the **top numbers** (nucleon number) and the **bottom numbers** (charge).

WORKED EXAMPLE: BALANCING EQUATIONS

Alpha decay of radon-222: ${}^{222}_{86}Rn \rightarrow {}^{218}_{84}Po + {}^{4}_{2}He$. Top: 222 = 218 + 4 ✓; bottom: 86 = 84 + 2 ✓.
An alpha particle hitting beryllium: ${}^{4}_{2}He + {}^{9}_{4}Be \rightarrow {}^{12}_{6}C + {}^{1}_{0}n$. Top: 4 + 9 = 12 + 1 ✓; bottom: 2 + 4 = 6 + 0 ✓.

MARK-SCHEME LANGUAGE

Often asked as: MCQ counting protons and neutrons from the symbol (e.g. A = 24, Z = 10 → 14 neutrons); completing a nuclear equation.

COMMON MISTAKE

Swapping proton and neutron numbers: every MCQ has the reversed pair as a distractor.
Reading "nucleon number" as "neutron number". Nucleons are protons **and** neutrons.

SPEC 112 Alpha scattering and the nuclear model of the atom

WHAT THIS SPEC POINT ASKS

Understand how large-angle alpha-particle scattering gave evidence for a nuclear atom, and how ideas about atomic structure have changed.

112 The experiment (Geiger and Marsden, explained by Rutherford, 1909–1911)

- A narrow beam of **alpha particles** was aimed at a very **thin gold foil** in an **evacuated** chamber.
- A detector (originally a zinc sulfide screen viewed through a microscope, giving tiny flashes) was moved round the foil to count alpha particles at each scattering angle.

Observation	Conclusion
Most alpha particles went **straight through**, undeflected	the atom is **mostly empty space**
Some were deflected through **small angles**	there is a **concentration of (positive) charge** that repels the alphas
A **very few** (about 1 in 8000) were deflected by **more than 90°**, some straight back	the positive charge and **most of the mass** are concentrated in a **tiny region**: the **nucleus**

- The count rate falls very steeply as the scattering angle increases.

112 Why the plum-pudding model failed

Thomson's model (1904) spread positive charge and mass **evenly** through the atom, with electrons dotted in it. That diffuse charge could only produce **small** deflections. Only a small, dense, charged nucleus can produce a force strong enough to turn an alpha particle back.

112 Details of the set-up

- **Vacuum:** so alpha particles are not absorbed or scattered by **air molecules** (their range in air is only a few cm).
- **Thin foil:** so each alpha is scattered by **one** nucleus at most (no multiple deflections) and is not absorbed.
- **Gold:** can be hammered extremely thin; its large nuclear charge (79e) gives big deflections.
- **Alpha particles**, not beta or gamma: beta particles are much less massive and are deflected by the atomic **electrons**; gamma rays are **uncharged**, so no electrostatic deflection. Both are also too penetrating.

112 How the model of the atom changed

Date	Model	Key idea
~1800	Dalton	atoms are tiny indivisible spheres
1897–1904	Thomson	discovered the electron; plum-pudding model
1911	Rutherford	small, dense, positive nucleus; electrons outside
1913	Bohr	electrons in fixed energy levels (explains line spectra)
1932	Chadwick	discovered the neutron, inside the nucleus
1960s–70s	high-energy electron scattering	protons and neutrons are made of quarks

MARK-SCHEME LANGUAGE

The six-mark answer pairs each observation with its conclusion:

- "most went straight through (undeviated), so the atom is mostly **empty space**"
- "a few deflected through small angles, so there is a **concentration of charge**"
- "very few deflected by **more than 90°**, so most of the **mass** is concentrated in a very **small** region (nucleus)"
- plum-pudding "predicted only small deflections"; the new model has "a very small nucleus containing almost all the mass and charged".

"Thin foil so alpha particles are less likely to undergo **multiple deflections**" or "less likely to be absorbed".

Often asked as: 6-mark description and significance; why vacuum, thin foil, gold; why not beta or gamma; MCQ on which statement was **not** a conclusion.

COMMON MISTAKE

Writing "deflected through angles **less than** 90°" for the large-angle evidence. It is **more than 90°**.
Listing observations without the conclusion each supports.
Saying "the nucleus contains protons and neutrons" was a conclusion. Neutrons were not discovered until 1932.
"Reflected off the nucleus": say **repelled** by the positive nucleus.

BEYOND THE SPEC, BUT EXAMINED

- **Closest approach** gives an upper limit for the nuclear radius: set the alpha's kinetic energy equal to the electrical potential energy, $E_k = (2e)(79e)/4\pi\varepsilon_0 r$ (see 98; WPH14 Jun 2022, Oct 2023).
- **Momentum in scattering:** an alpha bouncing off a light nucleus (or proton) is a 2-D collision problem (WPH14 Jun 2023).

SPEC 113 Thermionic emission and accelerating electrons

WHAT THIS SPEC POINT ASKS

Electrons are released by thermionic emission and can be accelerated by electric and magnetic fields.

113 Thermionic emission

- A metal **filament** (or cathode) is **heated** by passing a current through it.
- Some of the free (conduction) electrons gain enough energy to **escape from the metal surface**. This is **thermionic emission**.
- It must happen in a **vacuum**, so the electrons do not collide with gas molecules.
- A hotter filament (bigger heater current or heater p.d.) emits **more** electrons per second, giving a **larger beam current**.

113 Accelerating them: the electron gun

- A positive **anode** near the cathode creates an **electric field** that accelerates the electrons. Many pass through a hole in the anode as a beam.
- The work done on each electron by the accelerating p.d. V becomes kinetic energy:

$$eV = \tfrac{1}{2}mv^2$$

- e = 1.60×10^{-19} C, V = accelerating p.d. (V), m = 9.11×10^{-31} kg, v = final speed (m s^{-1}).
- A larger anode p.d. gives a **faster** beam; it does not change how many electrons are emitted.
- A **magnetic** field cannot change the electrons' speed: its force is perpendicular to their velocity, so it only changes their **direction** (steering and focusing beams, or bending them into circles).

WORKED EXAMPLE: SPEED FROM AN ELECTRON GUN

Electrons are accelerated through 2.0 kV.
$E_k = eV = 1.60 \times 10^{-19} \times 2.0 \times 10^{3} = 3.2 \times 10^{-16}$ J
$v = \sqrt{(2E_k/m)} = \sqrt{(2 \times 3.2 \times 10^{-16} \div 9.11 \times 10^{-31})} =$ **2.7×10^{7} m s^{-1}**
That is 9% of c, so the non-relativistic formula is still fine.

MARK-SCHEME LANGUAGE

- Two marks: "the (metal) **filament is heated** (by the current)" and "the electrons are released by **thermionic emission**" (or "electrons gain enough energy to leave the surface").
- DO NOT accept "thermal effect".
- "Use of Ek = QV" and "use of $E_k = \tfrac{1}{2}mv^2$"; some questions require at least 3 s.f.

Often asked as: name and describe the emission; speed of electrons from a given p.d.; MCQ on how to increase the beam current.

COMMON MISTAKE

Not naming thermionic emission, or not saying what is heated.
Saying the accelerating p.d. increases the number of electrons emitted.
Thinking a magnetic field can speed electrons up.

SPEC 114 Particle accelerators and detectors

WHAT THIS SPEC POINT ASKS

Understand the roles of electric and magnetic fields in linear accelerators, cyclotrons and particle detectors (ionisation and deflection only).

114 The linear accelerator (linac)

- A line of hollow metal **drift tubes**, connected alternately to the two terminals of a high-frequency **alternating p.d.** of **constant frequency**, all inside a vacuum.
- Particles are accelerated by the **electric field in the gaps** between tubes.
- Inside a tube there is **no electric field** (the tube shields it), so the particle **drifts at constant speed**.
- The p.d. **reverses while the particle is inside a tube**, so when it reaches the next gap the field there is again in the accelerating direction.
- The time spent in each tube is the same (half a period of the a.c.), but the particle gets faster, so **each tube must be longer** than the last (length = v × T/2).
- For electrons close to the speed of light the speed hardly increases any more, so the tubes stop getting longer (the energy still increases).

WORKED EXAMPLE: DRIFT-TUBE LENGTH

A proton moving at 2.0×10^6 m s^{-1} enters a tube; the a.c. frequency is 25 MHz.
Time in the tube = T/2 = $1 \div (2 \times 25 \times 10^6) = 2.0 \times 10^{-8}$ s
Length = vt = $2.0 \times 10^6 \times 2.0 \times 10^{-8}$ = **0.040 m**

114 The cyclotron

- Two hollow D-shaped electrodes (**dees**) in a vacuum, with a uniform **magnetic field perpendicular** to the dees and an **alternating p.d.** across the gap between them.
- **Electric field** (in the gap only): **accelerates** the particles each time they cross the gap.
- **Magnetic field**: exerts a force **perpendicular to the velocity**, which provides the **centripetal force**, so inside each dee the particle moves in a **semicircle** (no change of speed).
- After each crossing the particle is faster, so the radius (r = mv/BQ) is bigger: the path is an outward **spiral**. It leaves at the edge.
- Time for a semicircle: T/2 = πm/BQ. This **does not depend on speed or radius**, so the p.d. can switch at a **constant frequency** f = BQ/2πm, reversing while the particle is inside a dee.
- At speeds approaching c the particle takes longer per revolution (a relativistic effect), falls out of step with the p.d., and the cyclotron stops working.

WORKED EXAMPLE: A PROTON CYCLOTRON

B = 0.80 T, dee radius 0.40 m, proton.
Period T = $2\pi m/BQ = 2\pi \times 1.67 \times 10^{-27} \div (0.80 \times 1.60 \times 10^{-19}) = 8.2 \times 10^{-8}$ s, so f = 1/T = **12 MHz**
Exit speed: v = $BQr/m = 0.80 \times 1.60 \times 10^{-19} \times 0.40 \div 1.67 \times 10^{-27} = 3.1 \times 10^7$ m s^{-1}
Ek = $\frac{1}{2}mv^2 = 7.8 \times 10^{-13}$ J = **4.9 MeV**
With 20 kV across the gap, each crossing adds 20 keV, so about 245 crossings, or 123 turns (two crossings per turn).

114 Detectors

- **Ionisation:** a fast charged particle knocks electrons out of atoms along its path. The ions are made visible as a **track** (droplets in a cloud chamber, bubbles in a bubble chamber) or detected as electrical signals (wire and spark chambers, semiconductor detectors).
- **Neutral particles do not ionise**, so they leave **no track**. Their presence is inferred (e.g. from a gap, or from missing momentum).
- **Deflection:** a magnetic field across the detector bends the tracks. The **direction** of bending gives the **sign of the charge**; the **radius** gives the **momentum** (r = p/BQ).
- Heavily ionising particles (alphas, slow particles) leave thick, short tracks; fast electrons leave thin tracks.

MARK-SCHEME LANGUAGE

Cyclotron (the five points):

- "an **electric field** (from the p.d.) **between the dees** accelerates the particles"
- "the magnetic field is **perpendicular** to the velocity, so the **force is perpendicular** to the velocity"
- "this provides the **centripetal force**, so the path in a dee is a semicircle"
- "the p.d. switches polarity **when the particle is inside a dee**, so it is accelerated each time it crosses the gap"
- "constant frequency, because T = 2πm/BQ is **independent of speed** and radius"

Linac: "accelerated by the **electric field in the gaps**", "no field inside the tubes: constant speed", "a.c. of constant frequency, polarity changes while the particle is inside a tube", "time in each tube constant, so tubes get **longer** as speed increases", "near c the speed is almost constant, so tubes stop getting longer".

Detectors: "only charged particles leave tracks", "tracks formed by ionisation", "radius decreasing means the particle is losing energy".

Often asked as: 6-mark explanation of a linac or cyclotron, or a comparison of the two; cyclotron period, frequency or exit speed; sketch the spiral path; identify neutral particles from gaps in tracks.

COMMON MISTAKE

Saying the magnetic field accelerates the particles (it only changes their direction).
Saying "the p.d. accelerates them" without mentioning an **electric field**, or not saying **where** the field is.
Not saying **when** the polarity changes.
Using one dee for a whole period: the time in one dee is **T/2**.
Describing the cyclotron without explaining why the frequency can be constant.

BEYOND THE SPEC, BUT EXAMINED

- Deriving T = 2πm/BQ from r = p/BQ and v = 2πr/T (WPH14 Jan 2020).
- Counting gap crossings: energy per crossing = QV; two crossings per revolution (WPH14 Jan 2020, Jun 2025).
- **Synchrotron radiation:** charged particles moving in circles emit radiation and lose energy, so the radius shrinks (WPH14 Jun 2024 MCQ).

SPEC 115

Radius of a charged particle's path: r = p/BQ

WHAT THIS SPEC POINT ASKS

Derive and use r = p/BQ for a charged particle moving at right angles to a magnetic field.

115 Derivation

The magnetic force is perpendicular to the velocity, so it provides the centripetal force:

$$BQv = mv^2/r \quad \Rightarrow \quad r = mv/BQ = p/BQ$$

- r = radius (m), p = momentum (kg m s^{-1}), B = flux density (T), Q = charge (C).
- **Faster or heavier** particles (more momentum) make **bigger** circles. **Stronger fields** or **larger charges** make **smaller** circles.
- A particle losing energy (e.g. by ionising) slows down, so its radius **decreases**: a **spiral** track.
- The time for one orbit, T = 2πm/BQ, does not depend on the speed.

WORKED EXAMPLE: ELECTRON BEAM IN A FIELD

Electrons from a 2.0 kV gun ($v = 2.65 \times 10^7$ m s^{-1}, from 113) enter a 2.5 mT field at right angles.
$p = mv = 9.11 \times 10^{-31} \times 2.65 \times 10^7 = 2.41 \times 10^{-23}$ kg m s^{-1}
$r = p/BQ = 2.41 \times 10^{-23} \div (2.5 \times 10^{-3} \times 1.60 \times 10^{-19}) =$ **0.060 m**

WORKED EXAMPLE: AN ALPHA PARTICLE

A 5.0 MeV alpha particle ($m = 6.64 \times 10^{-27}$ kg, $Q = 2e$) moves at right angles to a 1.2 T field.
$E_k = 5.0 \times 10^6 \times 1.60 \times 10^{-19} = 8.0 \times 10^{-13}$ J; $p = \sqrt{(2mE_k)} = \sqrt{(2 \times 6.64 \times 10^{-27} \times 8.0 \times 10^{-13})} = 1.03 \times 10^{-19}$ kg m s^{-1}
$r = 1.03 \times 10^{-19} \div (1.2 \times 3.20 \times 10^{-19}) =$ **0.27 m**

MARK-SCHEME LANGUAGE

- Derivation: "equates $F = mv^2/r$ and $F = BQv$", "substitutes $p = mv$ with suitable algebra to arrive at $r = p/BQ$".
- Track photographs: "determines the radius from measurements", "**applies the scale**".
- Momentum units: N s or kg m s^{-1}, or GeV/c.

Often asked as: derive $r = p/BQ$; find B or p from a radius; radius from a photograph with a scale; MCQ (four times the mass, twice the charge, same speed → radius doubles).

COMMON MISTAKE

Using the diameter as the radius.
Using the electron mass for a proton.
For an alpha particle, using Q = e instead of 2e.

BEYOND THE SPEC, BUT EXAMINED

- **Very fast electrons:** $p = mv$ no longer works. Use **$E = pc$** (E in J) to get the momentum (WPH14 Jun 2021).
- **Mass spectrometer:** ions of the same charge and speed but different mass follow circles of different radius, so isotopes separate (WPH14 Oct 2022).

SPEC 116

Conservation laws and particle tracks

WHAT THIS SPEC POINT ASKS

Apply conservation of charge, energy and momentum to particle interactions, and interpret particle tracks.

116 What is conserved in every interaction

- **Charge:** total charge before = total charge after.
- **Energy** (mass–energy): total of rest energies + kinetic energies before = total after. Kinetic energy can become mass and mass can become kinetic energy ($\Delta E = c^2\Delta m$).
- **Momentum:** vector sum before = vector sum after. If a particle at rest decays into two, they fly apart in **opposite** directions. A momentum vector diagram must **close**.

116 Reading a track photograph



(a) same speed: opposite charges curve opposite ways

(b) pair production: r shrinks as Ek falls

Field into the page. (a) At the same speed, positive and negative particles curve opposite ways (Fleming's left-hand rule); a neutral particle leaves no track and is not deflected. (b) A gamma photon (no track) makes an electron–positron pair: two tracks from one point, curving opposite ways, spiralling inwards as they lose energy.

What you see	What it tells you
Track curves	particle is charged; curvature direction + field direction (FLHR) → **sign** of charge
Large radius	large **momentum** (r = p/BQ)
Radius shrinks along the track (spiral)	particle is **losing energy** and slowing; it travels **towards** the tighter curl
Track bends more after passing through a plate	it lost energy in the plate, so it was travelling **into** the more curved part
Tracks appear from nowhere ("V" shape)	a **neutral** particle (no track) decayed there
Two tracks from one point, curving opposite ways	pair production, or decay into opposite charges
Track suddenly stops or kinks	the particle decayed (perhaps into neutral particles) or collided
Momentum before ≠ sum of visible momenta after	a **neutral** particle carried the missing momentum

WORKED EXAMPLE: FIELD DIRECTION FROM A TRACK

A positron moves to the right across the page and curves **upwards**. Which way is the field?
The positron is positive, so the second finger (current) points right. The thumb (force) points up the page (towards the centre of curvature). The first finger then points **into the page**: the field is into the page.
An electron in the same place would curve **downwards**.

MARK-SCHEME LANGUAGE

- Particle deduction (6-mark style): "rest mass-energy of the incoming particles **plus their kinetic energy** = total after"; "some of the kinetic energy becomes **mass** ($\Delta E = c^2\Delta m$)"; "the momentum of the incoming particle equals the **vector sum** of the products (x-components), and the y-components **cancel**".
- Momentum diagram: "vectors drawn **end to end**", "arrows", "three lines form a **closed triangle**, so momentum is conserved".
- "Radius **decreasing**, so the particle is slowing / losing energy".

Often asked as: identify a missing (neutral) particle using charge and momentum; deduce field direction or charge sign from curvature; explain why a neutral particle leaves no track; draw a momentum triangle for a decay.

COMMON MISTAKE

Saying "the curvature decreases" when you mean the **radius** decreases (curvature increases).
Drawing a momentum triangle with the wrong particle's momentum in it.
Forgetting that neutral particles are invisible, so a gap in the picture can be a real particle.

BEYOND THE SPEC, BUT EXAMINED

Using the gap at the start of a track, or the order of tracks, to decide which event happened first (WPH14 Oct 2020).

SPEC 117 Why high energies are needed to probe nucleons

WHAT THIS SPEC POINT ASKS

Understand why high energies are needed to investigate the structure of nucleons.

117 The resolution argument

- To see detail of a certain size, you need a probe whose **wavelength** is about that size or smaller (otherwise it just diffracts around the object).
- Moving particles have a **de Broglie wavelength**:

$$\lambda = h / p$$

- λ = wavelength (m), h = Planck constant = 6.63×10^{-34} J s, p = momentum (kg m s^{-1}).
- A nucleon is about **10^{-15} m** across. To resolve its structure, λ must be about 10^{-15} m or less, so the probe needs a **very large momentum**, and so a **very high energy**.
- **Electrons** are used as the probe: they are **leptons**, so they do not feel the strong force, and they are point-like, so they probe the charge inside cleanly.

WORKED EXAMPLE: ENERGY NEEDED

For $\lambda = 1.0 \times 10^{-15}$ m: $p = h/\lambda = 6.63 \times 10^{-34} \div 1.0 \times 10^{-15} =$ **6.6×10^{-19} kg m s^{-1}**
These electrons are highly relativistic, so $E \approx pc = 6.63 \times 10^{-19} \times 3.00 \times 10^{8} = 2.0 \times 10^{-10}$ J ≈ **1.2 GeV**.
(Using $\frac{1}{2}mv^2$ would give a speed far above c, which is impossible.)

117 A second, different reason: making new particles

- To **create** a massive particle, the collision must supply at least its rest energy, mc^2 (see 118).

- **Colliding beams** are better than a fixed target: two equal beams head-on have **zero total momentum**, so the products need not carry momentum or kinetic energy away, and **all** the energy is available to make new mass. With a fixed target, much of the energy stays as kinetic energy of the products.

MARK-SCHEME LANGUAGE

- Structure: "high energy → **large momentum** → **small (de Broglie) wavelength**", "wavelength comparable to the size of the nucleon ($\approx 10^{-15}$ m)", "better resolution".
- Colliding beams: "no resultant momentum before the collision, so the products can have little momentum and kinetic energy, so all the energy is available to form the mass of new particles".

Often asked as: why high energy is needed to probe structure; MCQ where "to create new particles" is the distractor; why colliding beams are used; is the energy enough to make a particle of a given mass.

COMMON MISTAKE

Answering a "structure" question with $E = mc^2$. That explains making particles, not resolving detail.
Not linking all three: energy → momentum → wavelength → resolution.

BEYOND THE SPEC, BUT EXAMINED

- Electron **diffraction** shows that electrons behave as waves: "an electron does not always behave as a particle" (WPH14 Jan 2023).
- **E = pc** for high-energy electrons (WPH14 Jun 2021).

SPEC 118

Creation and annihilation of matter and antimatter

WHAT THIS SPEC POINT ASKS

Use $\Delta E = c^2\Delta m$ for the creation and annihilation of matter and antimatter particles.

$$\Delta E = c^2 \Delta m$$

- ΔE = energy (J), Δm = mass (kg), $c = 3.00 \times 10^8$ m s^{-1}. Mass and energy are interchangeable.
- **Rest energy** of a particle = mc^2 (energy it has because of its mass alone).

118 Annihilation

- A particle meets its **antiparticle**: both disappear and their total energy (rest energy + kinetic energy) becomes **gamma photons**, usually **two**.
- If they were at rest, total momentum was zero, so the two photons move in **opposite directions** with **equal energies**. Each photon then carries the rest energy of **one** particle: $E = mc^2$.
- One photon alone cannot be produced, because it could not have zero momentum.

118 Pair production

- A high-energy **photon** turns into a **particle–antiparticle pair** (e.g. e^- and e^+).
- Minimum photon energy = **$2mc^2$** (the rest energy of both particles). Any extra becomes their kinetic energy.
- It happens near a **nucleus**, which recoils to conserve momentum.

WORKED EXAMPLE: ELECTRON–POSITRON ANNIHILATION

Rest energy of an electron: $E = mc^2 = 9.11 \times 10^{-31} \times (3.00 \times 10^8)^2 = 8.20 \times 10^{-14}$ J = 0.51 MeV.
Each of the two photons has $E = 8.20 \times 10^{-14}$ J, so $\lambda = hc/E = 6.63 \times 10^{-34} \times 3.00 \times 10^8 \div 8.20 \times 10^{-14} =$ **2.4×10^{-12} m** (gamma rays).

WORKED EXAMPLE: PAIR PRODUCTION

Minimum photon energy for an e^-e^+ pair = 2 × 0.511 MeV = 1.02 MeV.
A 1.50 MeV photon makes a pair: kinetic energy shared by the pair = 1.50 − 1.02 = **0.48 MeV**.

WORKED EXAMPLE: IS THERE ENOUGH ENERGY?

A collision makes 2.9 GeV available. Can it produce a proton–antiproton pair?
Rest energy needed = $2 \times 1.67 \times 10^{-27} \times (3.00 \times 10^8)^2 = 3.0 \times 10^{-10}$ J = 1.88 GeV.
1.88 GeV < 2.9 GeV, so **yes**; the remaining 1.0 GeV becomes kinetic energy.

MARK-SCHEME LANGUAGE

- "Use of $\Delta E = c^2\Delta m$", "conversion factor for eV", then a **comparison with a conclusion**: "1.88 GeV < 2.9 GeV so yes".
- Kinetic energy of products = energy available − rest energy of the products.
- Compare energies, not speeds.

Often asked as: mass of a particle in kg or GeV/c^2; photon energy or frequency from annihilation or pair production; energy left over as kinetic energy; is a process possible.

COMMON MISTAKE

Not squaring c, or rearranging $m = E/c^2$ wrongly.
Forgetting the "M" (10^6) or "G" (10^9).
Forgetting to subtract the rest energy of the products, or to add the initial kinetic energy of the colliding particles.
Giving each annihilation photon the rest energy of **both** particles.

BEYOND THE SPEC, BUT EXAMINED

- Finding the photon **frequency** from $E = hf$ in pair production (WPH14 Jun 2021).
- Comparing a calculated particle mass with a table value as a percentage difference (WPH14 Jun 2025).

SPEC 119

Energy and mass units: MeV, GeV, MeV/c^2, GeV/c^2

WHAT THIS SPEC POINT ASKS

Use MeV and GeV for energy and MeV/c^2, GeV/c^2 for mass, and convert between these and SI units.

119 The electronvolt

- **1 eV** is the energy gained by an electron accelerated through a p.d. of 1 V: 1 eV = 1.60×10^{-19} J.
- 1 keV = 10^3 eV; **1 MeV = 10^6 eV = 1.60×10^{-13} J**; **1 GeV = 10^9 eV = 1.60×10^{-10} J**.
- eV → J: **multiply** by 1.60×10^{-19}. J → eV: **divide**.

119 Mass in MeV/c^2 and GeV/c^2

- From $m = E/c^2$, a mass can be quoted as an energy divided by c^2.
- 1 MeV/c^2 = 1.60×10^{-13} J ÷ $(3.00 \times 10^8)^2$ = **1.78×10^{-30} kg**. 1 GeV/c^2 = 1.78×10^{-27} kg.
- **Momentum** in GeV/c: because GeV/c^2 × c = GeV/c, "GeV/c" is a unit of momentum.

Particle	Mass in kg	Mass in MeV/c^2
electron	9.11×10^{-31}	0.511
proton	1.67×10^{-27}	938 (≈ 0.94 GeV/c^2)
neutron	1.67×10^{-27}	940

WORKED EXAMPLE: CONVERSIONS BOTH WAYS

Proton, kg → MeV/c^2: $E = mc^2 = 1.67 \times 10^{-27} \times (3.00 \times 10^8)^2 = 1.503 \times 10^{-10}$ J; ÷ 1.60×10^{-13} = **939 MeV**, so m ≈ 939 MeV/c^2 (the accepted 938 MeV/c^2 comes from more precise constants).
Top quark, GeV/c^2 → kg: 173 GeV = $173 \times 10^9 \times 1.60 \times 10^{-19} = 2.77 \times 10^{-8}$ J; $m = E/c^2 = 2.77 \times 10^{-8}$ ÷ $(3.00 \times 10^8)^2$ = **3.08×10^{-25} kg**.

MARK-SCHEME LANGUAGE

- "Conversion factor for eV" is its own mark.
- "GeV/c has the same unit as momentum, because GeV/c^2 × c = GeV/c".

Often asked as: MCQ mass in kg from MeV/c^2; identifying GeV s^{-1} as a unit of power; converting a particle's mass for a comparison.

COMMON MISTAKE

Neglecting the M or G prefix.
Multiplying by c^2 when you should divide ($E = mc^2$ gives energy from mass; mass = E/c^2).

SPEC 120 Relativistic increase in particle lifetime

WHAT THIS SPEC POINT ASKS

Understand when the relativistic increase in a particle's lifetime matters (no relativistic equations needed).

120 The muon puzzle

- **Muons** are made by cosmic rays hitting the upper atmosphere, about 10 km up. At rest their mean lifetime is about **2.2 µs**.
- Even at 0.98c, a muon would travel only $s = vt = 0.98 \times 3.00 \times 10^8 \times 2.2 \times 10^{-6} \approx$ **650 m** in that time.
- Yet large numbers of muons **reach the ground**.
- Explanation: muons are moving **close to the speed of light**. For an **observer on Earth**, the muon's **lifetime is longer** (time dilation), so it travels much further before decaying.

120 When it matters

- Relativistic effects are tiny at everyday speeds. They become **significant** only at speeds that are a large fraction of c (roughly above about 0.1c, and large near c).
- In accelerators, unstable particles moving near c live longer in the **laboratory frame**, so they travel further and leave longer tracks before decaying.

- The same effects explain why $\frac{1}{2}mv^2$ fails for very fast particles, and why a cyclotron falls out of step at high speeds.

WORKED EXAMPLE: CLASSICAL DISTANCE

A pion with lifetime 2.6×10^{-8} s moves at 0.95c. Classically: $s = 0.95 \times 3.00 \times 10^8 \times 2.6 \times 10^{-8}$ = **7.4 m**. If it is detected 50 m away, its lifetime in the lab frame must be longer than 2.6×10^{-8} s: a relativistic effect.

MARK-SCHEME LANGUAGE

- "Muons travelling **close to the speed of light**" → "**relativistic** effect **increases** the particle's lifetime (as measured by the **observer**)" → "so it travels further than expected".
- "Time **dilation**" (do not accept "dilution").

Often asked as: 2-mark explanation of why a particle is detected far from where it was made; $s = vt$ to show the classical distance is too short; MCQ (lifetime increases, rate of decay decreases).

COMMON MISTAKE

Mentioning time dilation but not saying what happens to the **lifetime**.
Saying the particle's own lifetime changes: it is the lifetime measured by the **observer** that is longer.

SPEC 121

The standard quark-lepton model

WHAT THIS SPEC POINT ASKS

Know that particles can be classified as baryons (three quarks), mesons (quark + antiquark), leptons (fundamental) and photons, and that the model's symmetry predicted the top quark.

121 Fundamental particles

- **Quarks** come in six flavours. They are never found on their own, only inside **hadrons**. They have **fractional** charges.
- **Leptons** are **fundamental**: they are not made of anything smaller. They do **not** feel the strong force. The electron, muon, tau and their three neutrinos are leptons.
- The **photon** is the particle (quantum) of electromagnetic radiation and carries the electromagnetic force. It has no charge and no mass, and is neither a hadron nor a lepton.

121 Hadrons (made of quarks; feel the strong force)

- **Baryons** = **three quarks** (antibaryons = three antiquarks). Examples: proton **uud** (charge $+\frac{2}{3} + \frac{2}{3} - \frac{1}{3} = +1$), neutron **udd** ($+\frac{2}{3} - \frac{1}{3} - \frac{1}{3} = 0$).
- **Mesons** = a **quark and an antiquark**. Examples: $\pi^+ = u\bar{d}$ ($+\frac{2}{3} + \frac{1}{3} = +1$), $\pi^- = d\bar{u}$ ($-\frac{1}{3} - \frac{2}{3} = -1$), $\pi^0 = u\bar{u}$ or $d\bar{d}$.
- Every hadron has a **whole-number** charge.

121 Quantum numbers

Particle	Symbol	Charge / e	Baryon number B	Lepton number L
up, charm, top quarks	u, c, t	+⅔	+⅓	0
down, strange, bottom quarks	d, s, b	−⅓	+⅓	0
antiquarks	$\bar{u}$, $\bar{d}$ …	opposite: −⅔ or +⅓	−⅓	0
proton	p (uud)	+1	+1	0
neutron	n (udd)	0	+1	0
antiproton	$\bar{p}$ ($\bar{u}\bar{u}\bar{d}$)	−1	−1	0
pions	π^+, π^0, π^-	+1, 0, −1	0	0
electron, muon, tau	e^-, μ^-, τ^-	−1	0	**+1**
neutrinos	ν_e, ν_μ, ν_τ	0	0	**+1**
positron, antimuon	e^+, μ^+	+1	0	−1
antineutrinos	$\bar{\nu}_e$, $\bar{\nu}_\mu$	0	0	−1
photon	γ	0	0	0

- Baryon number: each quark +⅓, so any baryon (3 quarks) has B = +1; mesons (⅓ − ⅓) have B = 0.
- Lepton number: every lepton **+1** (including the negatively charged electron), every antilepton **−1**.

121 Generations and the top quark

Generation	Quarks	Leptons
1st	up (u), down (d)	electron (e^-), electron neutrino (ν_e)
2nd	charm (c), strange (s)	muon (μ^-), muon neutrino (ν_μ)
3rd	top (t), bottom (b)	tau (τ^-), tau neutrino (ν_τ)

The model groups the particles into three generations, each with **two quarks and two leptons**. After the bottom quark was found (1977) and the tau lepton was known, the **symmetry** of the model demanded a sixth quark to pair with the bottom quark. The **top quark** was predicted, and was found in 1995 at Fermilab.

MARK-SCHEME LANGUAGE

- Baryon: "**3 quarks** (or 3 antiquarks)". Meson: "a **quark and an antiquark**" (ACCEPT $q\bar{q}$; DO NOT ACCEPT "quarks and antiquarks").
- Lepton: "fundamental / cannot be broken down", "L = ±1 and B = 0".
- Top quark: "a quark to match each lepton", "**by symmetry** of the standard model".
- "Mesons and baryons are **hadrons**."

Often asked as: definitions; build a meson or baryon from given quarks with the right charge; quark make-up of antiproton or pion; why the top quark was predicted; MCQ classification.

COMMON MISTAKE

Confusing a **muon** (a lepton) with a **meson**.
Writing "a baryon is made of quarks" without the number **three**.
Saying a neutron is "a proton plus an electron".
Giving a quark baryon number 0 (it is ⅓).

BEYOND THE SPEC, BUT EXAMINED

- Charm, strange, top and bottom quarks and their charges (WPH14 Jun 2023, Jan 2023).
- **Strangeness:** the s quark has strangeness −1 ($\bar{s}$ has +1); questions give it when needed, e.g. building a strange meson $s\bar{u}$ with charge −1 (WPH14 Jan 2024).
- The muon is a second-generation lepton, about 207 times the electron's mass (WPH14 Jan 2022).

SPEC 122

Particles and antiparticles

WHAT THIS SPEC POINT ASKS

Every particle has an antiparticle; use a particle's properties to deduce its antiparticle's, and the other way round.

122 The rule

An antiparticle has:

- the **same mass** (and so the same rest energy),
- the **opposite charge**,
- the **opposite baryon number** and **opposite lepton number** (and opposite quark flavours).

Particle	Antiparticle	What changes
electron e^- (Q −1, L +1)	positron e^+ (Q +1, L −1)	charge and L
proton p (uud, B +1)	antiproton $\bar{p}$ ($\bar{u}\bar{u}\bar{d}$, Q −1, B −1)	every quark → its antiquark
neutron n (udd)	antineutron $\bar{n}$ ($\bar{u}\bar{d}\bar{d}$)	still neutral, but B = −1, so a different particle
neutrino ν_e (L +1)	antineutrino $\bar{\nu}_e$ (L −1)	lepton number
π^+ ($u\bar{d}$)	π^- ($\bar{u}d$)	swap quark ↔ antiquark
photon γ	photon γ	it is its own antiparticle

- Naming: usually "anti-" plus the name and a bar over the symbol. The **positron** is the exception in name.
- To get an antiparticle's quark content, replace **each** quark with its antiquark.

WORKED EXAMPLE: DEDUCING AN ANTIPARTICLE

The Σ^+ baryon has quarks uus, charge +1, B = +1, L = 0, mass 1189 MeV/c^2.
Anti-Σ^+: quarks $\bar{u}\bar{u}\bar{s}$, charge $-\frac{2}{3} - \frac{2}{3} + \frac{1}{3} = \mathbf{-1}$, B = **−1**, L = 0, mass **1189 MeV/c^2**.

MARK-SCHEME LANGUAGE

- "**Same mass**, **equal and opposite charge**, **opposite lepton number**" (for a lepton) or "opposite baryon number" (for a baryon).
- "DO NOT ALLOW lepton number of an electron = −1 or charge of an electron = +1": the **electron has L = +1**.
- Pair creation: two Ω particles would break baryon conservation (0 → 2); an Ω and an anti-Ω conserve it (0 → 1 − 1 = 0).

Often asked as: state two properties of an antiparticle; quark make-up of antiparticles; MCQ comparing a particle and its antiparticle.

COMMON MISTAKE

Giving the electron lepton number −1 because it is negative. Lepton number has nothing to do with charge: e^- is **+1**, e^+ is **−1**.
Getting the positron or neutrino lepton numbers wrong: the most common error in a recent report.
Changing only some quarks when forming an antiparticle.

SPEC 123

Conservation of charge, baryon number and lepton number

WHAT THIS SPEC POINT ASKS

Use conservation of charge, baryon number and lepton number to decide whether a particle interaction is possible.

123 The rule

In **every** interaction or decay, these totals are the same before and after:

- **charge** Q
- **baryon number** B
- **lepton number** L

(Energy and momentum are also always conserved; see 116 and 118.) If any one of Q, B or L is not conserved, the interaction **cannot happen**.

Some questions give separate **electron** and **muon** lepton numbers. If so, each family's lepton number must be conserved separately. That is why π^+ decays to $\mu^+ + \nu_\mu$, not $\mu^+ + \nu_e$.

123 How to set out the answer

1. Write each particle's value **under** it, including the zeros.
2. Add each side.
3. State "total before = total after, so conserved" (or "≠, so not conserved") for **each** quantity, then the conclusion.

WORKED EXAMPLE: BETA-MINUS DECAY

$n \rightarrow p + e^- + \bar{\nu}_e$

Charge: $0 \rightarrow (+1) + (-1) + 0 = 0$ ✓

Baryon number: $1 \rightarrow 1 + 0 + 0 = 1$ ✓

Lepton number: $0 \rightarrow 0 + 1 + (-1) = 0$ ✓

All three are conserved, so the decay is possible. Without the antineutrino, L would be $0 \rightarrow 1$: **not** possible. This is why the antineutrino must exist.

WORKED EXAMPLE: AN IMPOSSIBLE REACTION

$p + p \rightarrow p + p + \bar{p}$?

Charge: $(+1) + (+1) = 2 \rightarrow (+1) + (+1) + (-1) = 1$ ✗

Baryon number: $1 + 1 = 2 \rightarrow 1 + 1 + (-1) = 1$ ✗

Not possible. The real reaction makes a **pair**: $p + p \rightarrow p + p + p + \bar{p}$ (charge $2 \rightarrow 2$, B $2 \rightarrow 2$ ✓).

MARK-SCHEME LANGUAGE

- "Lepton number: $0 + 0 \rightarrow 0 + 0 - 1 + 1$ (0 on each side, so conserved)": values for **every** particle, then the comparison.
- Marks come in pairs: "show the charge for all particles", then "total charge before = total charge after, so charge is conserved" (dependent on the first).
- Some questions want only two of the three laws ("MAX 2").

Often asked as: show that an interaction is (or is not) possible; identify which law it breaks; MCQ.

COMMON MISTAKE

"Baryon number is conserved because it is 2 before and 2 after" with no values for each particle: zero marks.

Writing "+1 + 1 + 0" but never showing the totals are equal.

Putting charge values in the lepton-number line.

Giving the neutron baryon number 0.

SPEC 124

Writing and interpreting particle equations

WHAT THIS SPEC POINT ASKS

Write and interpret particle equations using the relevant particle symbols.

124 Equations you should know

Process	Equation	Quark-level change
β^- decay	$n \rightarrow p + e^- + \bar{\nu}_e$	$d \rightarrow u + e^- + \bar{\nu}_e$
β^+ decay (in a nucleus)	$p \rightarrow n + e^+ + \nu_e$	$u \rightarrow d + e^+ + \nu_e$
electron capture	$p + e^- \rightarrow n + \nu_e$	$u + e^- \rightarrow d + \nu_e$
pair production	$\gamma \rightarrow e^- + e^+$	
annihilation	$e^- + e^+ \rightarrow \gamma + \gamma$	
pion decay	$\pi^+ \rightarrow \mu^+ + \nu_\mu$ and $\pi^- \rightarrow \mu^- + \bar{\nu}_\mu$	
alpha decay	${}^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}He$	

- In β^- decay a **lepton** (e^-) is made, so an **anti**lepton ($\bar{\nu}_e$) must be made too, keeping L = 0. In β^+ decay an antilepton (e^+) is balanced by a lepton (ν_e).
- Check every equation for charge, baryon number and lepton number (123), and for nuclear equations also the nucleon and proton numbers (111).

WORKED EXAMPLE: IDENTIFY THE UNKNOWN PARTICLE

$p + \bar{\nu}_e \rightarrow n + X$
Charge: $+1 + 0 = 0 + Q_X \rightarrow Q_X = \mathbf{+1}$
Baryon number: $1 + 0 = 1 + B_X \rightarrow B_X = \mathbf{0}$
Lepton number: $0 + (-1) = 0 + L_X \rightarrow L_X = \mathbf{-1}$
A positive antilepton with no baryon number: X is a **positron, e^+**.

MARK-SCHEME LANGUAGE

- Both particle **type** and particle/**anti**particle must be right: "there should be an **antineutrino** and not a neutrino".
- π^+ decay products: "$\mu^+ + \nu_\mu$"; the correct charged product is the **antimuon**.

Often asked as: write the equation for a decay given in words; complete an equation; identify X from conservation laws; MCQ on which equation is possible.

COMMON MISTAKE

Neutrino where an antineutrino is needed (or the reverse).
Charge not balanced, or the wrong sign on the muon.
Forgetting the gamma photons are two in annihilation at rest.

EXAM TIP

Do the lepton-number check last: it is the one that tells you whether you need a neutrino or an antineutrino.